

How AI Works

▼ Teacher resources

▼ Printing Options

These lessons supplement the [How AI Works](#) video series. Each lesson is paired with a single video from the series, or may deepen the concepts introduced in the videos. Lessons focus on creating an exciting first experience for students, where students engage in an activity or simulation and then reflect on the experience as a way to build understanding around AI and build empathy with those impacted by it. Some lessons focus more on the technology behind AI, while other lessons focus on the societal impacts and ethical considerations of AI to save your progress.

Supports for Teachers: You can learn more about bringing AI lessons to your classroom through our [AI 101 for Teachers series](#). AI 101 for Teachers is a free, foundational online professional learning series for any teacher and educator interested in the groundbreaking world of artificial intelligence (AI) and its transformative potential in education.

Implementation Options: Click one of the options below to expand and view more details.

- [I want to teach them all!](#)
- ["I want to teach a specific lesson or integrate into another class"](#)
- ["I Only Have a Day"](#)

Week 1	Lesson 1: Introduction to Machine Learning	Lesson 2: Computer Vision	Lesson 3: Neural Networks	Lesson 4: Chatbots and Large Language Models (LLMs)	Lesson 5: Generative Images
Week 2	Lesson 6: Algorithmic Bias	Lesson 7: Our AI Code of Ethics	Lesson 8: How AI Works In a Day		

Key	☐ Instructional Lesson	☒ Assessment	☒ Unplugged Lesson
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▼ Lesson 1: Introduction to Machine Learning

This lesson centers around the [How AI Works: What is Machine Learning?](#) video from the [How AI Works video series](#). Watch this video first before exploring the lesson plan.

In this lesson students are introduced to a form of artificial intelligence called machine learning and how they can use the Problem Solving Process to help train a robot to solve problems. They participate in three machine learning activities where a robot - AI Bot - is learning how to detect patterns in fish.

This lesson can be taught on its own, or as part of a 7-lesson sequence on How AI Works - [click here to view all lessons in this sequence](#).

🖥 1

Recognizing Fish

🖥 2

Recognizing Fish Features

▼ Lesson 2: Computer Vision

Sign in or provide your age to continue

This lesson centers around the [How AI Works: Computer Vision](#) video from the [How AI Works video series](#). Watch this video first before exploring the lesson plan.

Students learn how computer vision works. They first look at optical illusions to identify the features of the drawing that their eyes noticed. Students watch a video explaining computer vision and how a computer "sees". They design an algorithm that uses a network to decide what number the seven segment display is displaying. Finally, students test their algorithm.

Already have a Code.org account? Sign in so we can save your progress. [Sign in to Code.org](#)

This lesson can be taught on its own, or as part of a 7-lesson sequence on How AI Works - [click here to view all lessons in this sequence](#).

Privacy Policy

This lesson contains no levels.

▼ Lesson 3: Neural Networks

This lesson centers around the [How AI Works: Neural Networks](#) video from the [How AI Works video series](#). Watch this video first before exploring the lesson plan.

Students learn how neural networks work. They first discuss an example of an experience that recommends things to you. They then use a widget that recommends videos based on one person. Students watch a video explaining neural networks. They use an updated widget to adjust the weights of each person. Finally, students discuss the need for diverse perspectives when creating recommendation systems.

This lesson can be taught on its own, or as part of a 7-lesson sequence on How AI Works - [click here to view all lessons in this sequence](#).



1

WeTube Video Recommendations (Beta)

2

WeTube Video Recommendations (2.0)

▼ Lesson 4: Chatbots and Large Language Models (LLMs)

This lesson centers around the [How AI Works: Chatbots and Large Language Models](#) video from the [How AI Works video series](#). Watch this video first before exploring the lesson plan.

Large Language Models (LLMs) generate new text. The text LLMs generate looks like a human might have written it because large language models are built based on large bodies of text, such as Wikipedia. In this lesson, students learn what an LLM is and how it works, then use an LLM to co-create a text with AI. Finally, the class wraps up with a discussion about whether or not LLMs are intelligent or creative.

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1

Written Voice Simulator Widget



(Optional) Eliza Widget

▼ Lesson 5: Generative Images

Sign in or provide your age to continue

This lesson centers around the [How AI Works: Creativity and Imagination?](#) video from the [How AI Works video series](#). Watch this video first before exploring the lesson plan.

Diffusion models generate images. Diffusion AI converts an image to noise, and trains an AI to reverse the process. In this lesson, students learn how AI can generate images, then explore a diffusion AI widget. Finally, the class wraps up with a discussion about whether or not these models are creative.

This lesson can be taught on its own, or as part of a 7-lesson sequence on How AI Works - [click here to view all lessons in this sequence](#).

Privacy Policy



Noisy Image Widget



(Optional) GAN Guess Widget

▼ Lesson 6: Algorithmic Bias

This lesson centers around the [How AI Works: Equal Access and Algorithmic Bias](#) video from the [How AI Works video series](#). Watch this video first before exploring the lesson plan.

In this lesson, students will practice cropping images to uncover the bias underlying the Twitter cropping algorithm. Then, students will read and watch a video about the discovery of this biased algorithm. Finally, students will discuss ways to recognize and reduce bias along with analyzing Twitter's response to the allegations of bias in their cropping algorithm.

This lesson can be taught on its own, or as part of a 7-lesson sequence on How AI Works - [click here to view all lessons in this sequence](#).



Cropping Widget

▼ Lesson 7: Our AI Code of Ethics

This lesson centers around the [How AI Works: Privacy and the Future of Work](#) video from the [How AI Works video series](#). Watch this video first before exploring the lesson plan.

In small groups, students conduct research using articles and videos that expose ethical pitfalls in an Artificial Intelligence (AI) area of their choice. Afterward, there are options for how students synthesize what they've learned to make recommendations for how AI can be used in society: creating an AI Code of Ethics, or creating an AI Manifesto, or designing an "Imagine your World" artistic representation.

This lesson can be taught on its own, or as part of a 7-lesson sequence on How AI Works - [click here to view all lessons in this sequence](#).

This lesson contains no levels.

▼ Lesson 8: How AI Works In a Day

⚠ This lesson is intended for classrooms that want to show the entire **How AI Works video series** in a **single day**. It is not intended to be taught in sequence with the other lessons in this unit, which introduces each video one day at a time.

Sign in or provide your age to continue

Students follow along with each video by matching vocabulary from the video, then answering a reflection question about each video. The lesson plan and slides are very sparse and open-ended to allow for improvisation and customization to fit your classroom.

Already have a Code.org account? Sign in so we can save your progress.
This lesson contains no levels.

Provide your age below and click OK to continue.

or

Sign in to Code.org

[Privacy Policy](#)

Lesson 1: Introduction to Machine Learning

45 minutes

Overview

This lesson centers around the [How AI Works: What is Machine Learning?](#) video from the [How AI Works video series](#). Watch this video first before exploring the lesson plan.

In this lesson students are introduced to a form of artificial intelligence called machine learning and how they can use the Problem Solving Process to help train a robot to solve problems. They participate in three machine learning activities where a robot - AI Bot - is learning how to detect patterns in fish.

This lesson can be taught on its own, or as part of a 7-lesson sequence on How AI Works - [click here to view all lessons in this sequence](#).

Standards

Full Course Alignment

AI4K12 National Guidelines 2021

- ▶ **BI-3** - Computers can learn from data

CSTA K-12 Computer Science Standards (2017)

- ▶ **AP** - Algorithms & Programming
- ▶ **IC** - Impacts of Computing

Agenda

Before the Lesson

Warm Up (10 minutes)

Journal

Activity (30 minutes)

Level 1 - Recognizing Fish

Level 2 - Recognizing Fish Features

Level 3 - Recognizing Fish Expressions

Wrap Up (5 minutes)

Journal

After the Lesson

Teacher Survey

Additional Lessons

AI and Machine Learning Unit

Objectives

Students will be able to:

- Apply the Problem Solving Process to train a computer to solve a problem

Preparation

- Review the Code Studio levels before the lesson
- Print copies of the activity guide for each student

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- [Introduction to Machine Learning](#) - Slides [▼ Make a Copy](#)

For the students

- [Classifying Fish](#) - Activity Guide [▼ Make a Copy](#)
- [How AI Works - Training Data and Bias](#) - Video ([Download](#))
- [How AI Works - What is Machine Learning?](#) - Video ([Download](#))

Vocabulary

- **Machine Learning** - How computers recognize patterns and make decisions without being explicitly programmed

Teaching Guide

Before the Lesson

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Become a Verified Teacher: Lesson plans and levels have additional resources and answer keys for Verified Teachers, which is quick process that verifies your position at an educational institution. [Click here to complete a form](#) and you should have access to verified teacher resources in ~1 business day.

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Warm Up (10 minutes)

Journal

 **Prompt:** *What's an example of Artificial Intelligence (or AI) either in your personal life or that you've seen in a movie or book?*

Teaching Tip

What Is AI? At a basic level, artificial intelligence is when a computer program mimics the intelligence of a human being. This can appear as solving a problem, engaging in conversations, displaying emotions, and many other forms. Students may gravitate towards examples of artificial intelligence whose form also mimics a human being, such as interactive robots or Siri or having a conversation with ChatGPT, but also emphasize examples that broader and more embedded in tools we use frequently, such as social media algorithms or streaming recommendation algorithms.

Shortly after this prompt, students will watch a video that further defines AI and the focus of this unit, Machine Learning.

Have students brainstorm silently on their own, then share with their neighbors, and finally share with the whole class.

 **Discussion Goal:** Try to surface any personal connections students may already have with AI. Students may come up with examples from their personal lives, such as recommendation systems or facial recognition. Or they may think of examples in the media, such as the robots in movies like Wall-E or personal assistants like Siri. Students may also come up with examples that aren't strictly AI - for now, add them to the list anyway. Keep track of any suggestions that students surface without validating them as right or wrong - let students brainstorm freely first.

 **Display:** Display the slide with the large venn diagram that includes AI and many applications.

Remarks

Artificial Intelligence is used in a lot of different places in our lives - from facial recognition in our phones to personal recommendations when we browse the web, and even in driverless cars. For the next few weeks, we're going to focus on a specific type of artificial intelligence called machine learning.

 **Video:** Show students the [How AI Works - What is Machine Learning?](#) video in the slides

Teaching Tip

Videos are used throughout the curriculum to spark discussions, supplement key concepts with additional explanations and examples, and expose students to the various roles and backgrounds of individuals in computer science.

While interacting with the video, turn on closed captioning so students can also read along as they watch.

To encourage active engagement and reflection, use one or more of the strategies discussed in the [Guide to Curriculum Videos](#).

 **Vocabulary:** Display the following vocabulary

- **Machine Learning:** How computers recognize patterns and make decisions without being explicitly programmed

Remarks

Machine learning helps us solve important problems in society. In the next few weeks, we'll look at how we can create our own machine learning apps to solve problems. To help accomplish our goals, we will use the Problem Solving Process - Define, Prepare, Try, Reflect, and always Empathize. Let's take a look at how these steps appear for Machine Learning in particular

 **Display:** Show the slide with the Problem Solving Process. Read through the additional lines that represent how the process connects to Machine Learning.

Activity (30 minutes)

Remarks

In today's activity, we're going to use machine learning to help a robot clean up the ocean. To do this, we need to teach the robot to recognize fish so it will clean up anything that isn't a fish. We'll give it lots of examples of fish, give it time to learn from those examples, and then see how well it does in cleaning up the ocean.

Teaching Tip

Real-World Influence: This type of technology is being used in the real-world by groups such as [The Ocean Cleanup](#), which have used artificial intelligence and machine learning to train robots to recognize plastics in the water. You could consider showing this short video to the class as a way to further motivate today's activity: [How AI Helps Clean Oceans From Plastics](#)

 **Distribute:** Pass out the Activity Guide for this lesson

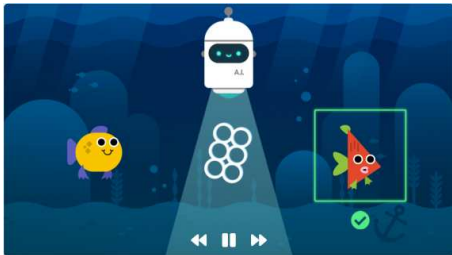
Level 1 - Recognizing Fish

 **Code Studio:** In front of the classroom, navigate to Level 1 in Code Studio - Recognizing Fish.

☑ **Do This:** As a class, go through this level which guides you through the process of training A.I. Bot to recognize fish. As you do, fill in the first part of the activity guide as a class. Consider the following strategies at different stages of the level:

💡 Teaching Tip

A.I. Bot and Pronouns: These levels purposefully use A.I. Bot's full name and avoid gendering A.I. Bot as "he" or "she". Even though some AI systems take on personified gendered roles - such as Siri or Jarvis - actual machine learning programs don't have genders because they're just computer programs. Model this same behavior with students by referring to A.I. Bot by its full name or using "it" as its pronoun.

Activity	As a Class
Training Screen 	• Informational text will frequently pop up on the screen. Consider asking students to read the prompts aloud to the class, possibly in their best robot voice, for a little fun • Ask students to use thumbs up / thumbs down to decide as a class whether something is a fish • At 30 images, the level will ask if you would like to continue or keep training. Ask students to help with this decision and follow their lead - either continue training, or go to the next screen. • Before continuing, have students record on their activity guide how many images they used for their training data
Testing Screen 	• A.I. Bot will probably misidentify a few objects. Make sure students notice this, and ask them to individually consider why they think this might be happening. This ties in to the discussion at the end of this level.
Evaluation Screen 	• Have students count the number of fish that it identified correctly. Record this on the activity guide. • It's okay if A.I. Bot misidentifies some trash as fish - this sets the stage for the discussion.

🗣️ **Discuss:** If we wanted A.I. Bot to become better at recognizing fish, how do you think we could help it do that?

💡 **Discussion Goal:** Help students understand that A.I. Bot learns better when it has more examples. The best way we can improve its ability to recognize fish is by giving it more examples to look at. Students may also make the connection that this is how humans learn, especially young children - the more examples we have, the better we learn.

Remarks

This is one example of how the Problem Solving Process and Machine Learning can be used to solve a problem. We prepared our data, used it to train A.I. Bot, then reflected on the results and decided what to do next. In this next level, you will follow this process and train A.I. Bot to detect certain kinds of fish - like red fish or blue fish or triangle fish.

Level 2 - Recognizing Fish Features

 **Code Studio:** Have students navigate to Code Studio Level 2 - Recognizing Fish Features. This level lets students train A.I. Bot to recognize a fish by its color or body type.

2

Recognizing Fish Features

 **Do This:** Have students choose a word that they want to train A.I. Bot to recognize, and record this information on their Activity Guide. Students should progress through the level on their own, recording information on their activity guide.

 **Circulate:** Check that students are filling in their activity guide as they complete the stages. After briefly checking in with each student, complete this stage yourself in front of the room and pause at the last screen.

Teaching Tip

Activity Guides and Code Studio: Students may struggle initially to keep track of information on their activity guide as they complete levels in Code Studio. One way to help students think about this is similar to being a scientist performing a lab experiment: they are performing an experiment in Code Studio, and keeping track of their results in their Activity Guide.

 **Display:** As students reach the final stage, have them press the white Information icon in the upper-right corner. This displays the features that A.I. Bot is using to help make its decision. In the example here, A.I. Bot has learned that the color matters the most when making a decision and the eyes matter the least.

Have students record the most important and least important features on their activity guide.

 **Discuss:** *What features did A.I. Bot think were the most important? Are those the features you were expecting to be most important?*

 **Discussion Goal:** Students may find that A.I. Bot thinks some features are important when they're actually completely unrelated to their word. For example, thinking that the mouth of a fish is important when trying to determine if the fish is red. Students can verify this by looking for patterns in the fish that A.I. Bot has accepted - maybe all of their example fishes happened to have the same type of mouth and A.I. Bot mistakenly thought this was important.

Remarks

Some of you noticed that A.I. Bot was learning about parts of the fish that aren't actually important, like the mouth or eyes or dorsal fin. This happens in real life too - machines can learn patterns we don't intend even with lots of data, which can cause A.I. Bot to make mistakes. For this situation, it's pretty easy to tell if A.I. Bot made a mistake - we can quickly see if a fish isn't actually red or circular. Let's try a slightly harder challenge - trying to recognize expressions!

Level 3 - Recognizing Fish Expressions

 **Code Studio:** Have students navigate to Code Studio Level 3 - Recognizing Fish Expressions. This level lets students train A.I. Bot to recognize a fish by its expression, such as “silly” or “serious” or “angry”.

3

Recognizing Expressions

 **Do This:** Have students choose an expression that they want to train A.I. Bot to recognize, and record this information on their Activity Guide. Students should progress through the level on their own, recording information on their activity guide. On the final screen, have students press the information icon and record the features that A.I. Bot learned were most important.

 **Circulate:** Check that students are filling in their activity guide as they complete the stages. Ask students what features they’re looking at to determine if a fish meets their criteria. Students will probably say they primarily use eyes and mouth to help determine expression, but when they click the Information icon on the final screen, A.I. Bot may be using additional features such as color and body to make its decision. Prompt students to think about how they feel about this and compare it to their personal experience - For example, if someone thought they were “silly” or “angry” primarily based on their clothing.

 **Display:** Show the slide of A.I. Bot learning how to identify fish as “Angry”. Have students discuss the prompt on the screen: *Looking at this screen, why do you think color appeared as the second most important feature?*

 **Discussion Goal:** Students should notice that even though the eyes and mouth tend to appear “angry” on every fish, there also appears to be a lot of purple fish on the screen. As a result, A.I. Bot may start to think that the color “purple” is another way to tell if a fish is angry or not.

 **Discuss:** *Do you think it’s okay to consider a fish “angry” by its color?*

 **Discussion Goal:** For the first part of the question, guide students to notice that there are a lot of purple fish on the screen so A.I. Bot might think that purple fish are more likely to be angry than other fish.

Teaching Tip

Job Interviews: If there is time available, consider showing students **Objective or Biased: On the questionable use of Artificial Intelligence for job applications**. This is a real-life example of a similar situation where unintended factors, such as wearing glasses or a headscarf, are influencing how an AI system rates job applicants. This can be a useful resource to make a connection between this fish activity and the real-world, but this resource is not designed for a middle-school audience and requires some adjustments and decisions for how to best present to your students.

The second part of the question is more open-ended. Students should explore their own ideas and feelings about whether this kind of labeling is appropriate based on the color of the fish. In particular, A.I. Bot has learned “purple fish are angry fish”, which students may have strong reactions to. They may describe similar situations outside of the classroom, such as moments where they’ve been judged by their appearance, gender, or race, to help explain their opinions. This question is designed to start a conversation and connect students’ experiences to the potential pitfalls of machine learning, and it’s okay if the discussion doesn’t come to a firm conclusion. You can let students know that they will continue to learn about and discuss these issues in the rest of the unit. Use the remarks below to help wrap up the discussion.

Remarks

Even with a small example like this, we see that machine learning can get into trouble and learn something untrue about a particular type of fish. Having more data and making sure it represents all types of fish can help solve this problem. These types of examples happen in real life too and can have serious consequences, such as whether or

not you receive medical care or get a job offer. As we learn how to use machine learning to solve problems, we need to always be thinking about: What is the impact, and who is being included or excluded?

Wrap Up (5 minutes)

🎥 **Video:** Show students the video in the slides [How AI Works - Training Data and Bias](#)

Journal

💬 **Prompt:** *What's a situation where it might be helpful to use machine learning to solve a problem?*

Encourage students to share with you on the way out the door, or to write their responses on a post-it note that can be displayed in the classroom. If you teach additional lessons within this unit, you can refer back to these post-its as these topics are covered in other lessons.

After the Lesson

Teacher Survey

We'd love to learn more about the folks teaching these lessons and the classroom experience. Please let us know in this [How AI Works Teacher Survey](#).

Additional Lessons

If you'd like to teach additional lessons from the How AI Works video series, [click here to explore additional resources](#)

AI and Machine Learning Unit

If you'd like to dive even deeper into AI and Machine Learning, consider exploring our 5-week unit on AI and Machine Learning. [Click here to view the unit and learn more](#)



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Lesson 2: Computer Vision

45 minutes

Overview

This lesson centers around the [How AI Works: Computer Vision](#) video from the [How AI Works video series](#). Watch this video first before exploring the lesson plan.

Students learn how computer vision works. They first look at optical illusions to identify the features of the drawing that their eyes noticed. Students watch a video explaining computer vision and how a computer "sees". They design an algorithm that uses a network to decide what number the seven segment display is displaying. Finally, students test their algorithm.

This lesson can be taught on its own, or as part of a 7-lesson sequence on How AI Works - [click here to view all lessons in this sequence](#).

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- ▶ **AP** - Algorithms & Programming
- ▶ **CS** - Computing Systems

Agenda

Before the Lesson

[Preparing to Teach a Code.org Lesson](#)

Warm Up (5 minutes)

[Optical Illusions](#)

Activity (35 minutes)

[Image Recognition Algorithm](#)

[Testing Our Algorithm](#)

Wrap up (5 minutes)

After the Lesson

[Teacher Survey](#)

[Additional Lessons](#)

[AI and Machine Learning Unit](#)

Objectives

Students will be able to:

- Describe how computer vision is built of layers that combine smaller components into a final prediction

Preparation

- Print copies of the Computer Vision Network activity guide for each group
- (Optional) Print copies of the Seven Segment Display Manipulative

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- [Computer Vision](#) - Slides
▼ Make a Copy
- [\[Example\] Computer Vision Network](#) - Exemplar
▼ Make a Copy

For the students

- [Computer Vision Network](#) - Activity Guide
▼ Make a Copy
- [How AI Works: How Computer Vision Works](#) - Video
- [Seven Segment Display Manipulative](#) - Activity Guide
▼ Make a Copy
- [What a Neural Network Sees](#) - Video

Vocabulary

- Algorithm** - a series of steps to solve a problem or make a decision
- Network** - A collection of items connected together. Following these

Teaching Guide

Before the Lesson

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Warm Up (5 minutes)

Optical Illusions

 **Display:** Show the image on the slide to students.

(Click to expand and see larger)

 **Prompt:** *What animal do you see in this image? Write your answer on a sheet of paper*

 **Discussion Goal:** After a brief pause, ask students to share out. Students will likely notice one of two animals: either a duck or a rabbit. Once both options are said, students may "switch" back and forth in how they see the image - sometimes it's a duck, sometimes it's a rabbit.

 **Prompt:** *If this were an argument and you wanted to convince someone else that your animal is correct, what would you say? What **features** in the image help your argument?*

💡 **Discussion Goal:** This prompt is meant to help students identify the features of the drawing that their eyes noticed to help make that decision. Both arguments will likely identify the black dot in the center as the eye of the animal. Duck arguments will identify the left feature as the bill or mouth of the duck, whereas rabbit arguments will identify those as the ears. Emphasize that students are using specific features to make their decision (ie: eye + ears + mouth = rabbit).

📺 **Display:** Show slide with additional optical illusions

💡 Teaching Tip ▲

Illusions: The illusions are:

- **The Dress:** Some folks see a blue dress; some folks see a gold dress; some folks see a variety of different colors. This picture went viral several years ago with debates over the color of the dress
- **Young Woman vs Older Woman:** Depending on how you view it, this image either looks like a young woman looking away off to the distance, or an older woman looking down. The chin and ears of the younger woman are the nose and eyes of the older woman.
- **Dogs and Bagels:** Not really an optical illusion - if you look closely, some of the images are of dogs and some are of bagels. At a quick glance, it's easy to miss which one is which.

🎤 **Remarks**

We use our eyes and sense of sight to identify objects, and sometimes an image with identical features can be interpreted in several different ways. But what about a computer - what would it "see" if it saw one of this pictures? Is the dress gold, or is it blue? Is this a young woman looking away, or an old woman looking down? Are these a bunch of dogs, or a bunch of bagels? Today we're going to talk about how computers can "see" and detect images

Activity (35 minutes)

Image Recognition Algorithm

🎥 **Video:** Show the [How AI Works: How Computer Vision Works](#) video.

💡 Teaching Tip ▲

Videos are used throughout the curriculum to spark discussions, supplement key concepts with additional explanations and examples, and expose students to the various roles and backgrounds of individuals in computer science.

While interacting with the video, turn on closed captioning so students can also read along as they watch.

To encourage active engagement and reflection, use one or more of the strategies discussed in the [Guide to Curriculum Videos](#).

💡 **Prompt:** How would you summarize how a computer "sees" something in an image?

💡 **Discussion Goal:** Listen for students describing how it breaks an image into parts, then combines them together in layers to create more complex shapes. Students may rely on the shape visuals from the video, or on the dog and muffin visuals.

✍️ **Vocabulary:** Display the slide with vocabulary for today's lesson

- **Algorithm** - a series of steps to solve a problem or make a decision
- **Network** - A collection of items connected together. Following these connections can represent the steps in an algorithm, like a flowchart.

Remarks

Today we're going to create our own image recognition **algorithm**. An algorithm is a series of steps to solve a problem or make a decision. We're going to create an algorithm that uses a **network** to make a decision, just like in the video

 **Display:** Show the slide with a scoreboard from a baseball game

Remarks

We're helping a sports arena improve its scoreboard. We want to train a camera to detect what the scores are so it can be broadcast to a radio receiver. For this to work, it needs to detect which number is displayed on the scoreboard. Each number is represented with one of these displays, which uses 7 segments to create images.

 **Prompt:** How would you represent your birthday using these numbers? Sketch your answer on a sheet of scratch paper

As students are sketching their responses, begin distributing the activity guide.

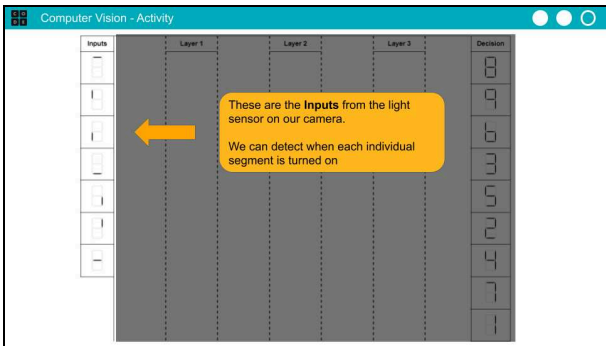
 **Distribute:** Pass out the **Computer Vision Network** activity guide

Teaching Tip

Going Big: Consider printing out copies of the 7-segment manipulatives and having students work on whiteboards, poster paper, rolls of butcher paper, or even setting down individual items and connecting them with string or yarn.

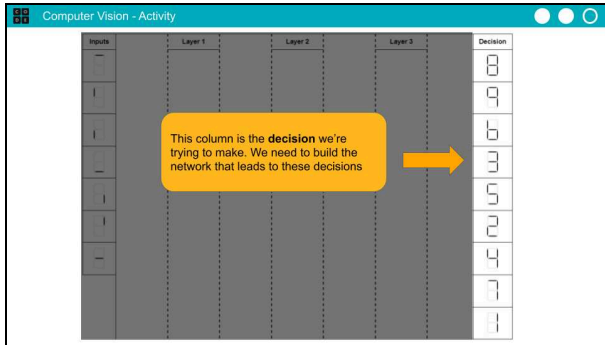
Group: Group students into pairs to work together

 **Display:** Show the slides giving an overview of this activity

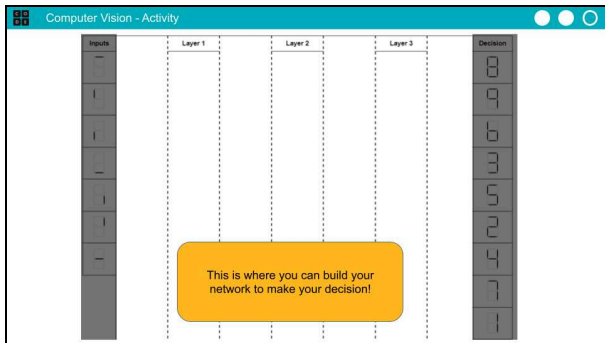
Slides	Say
	Say: The left side of this guide represents the inputs that your sensor can detect. It knows when each individual segment is turned on.

Slides

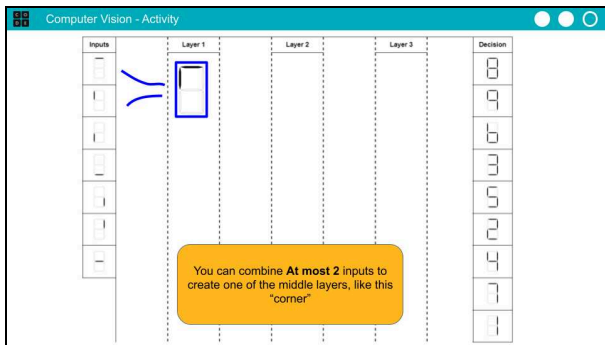
Say



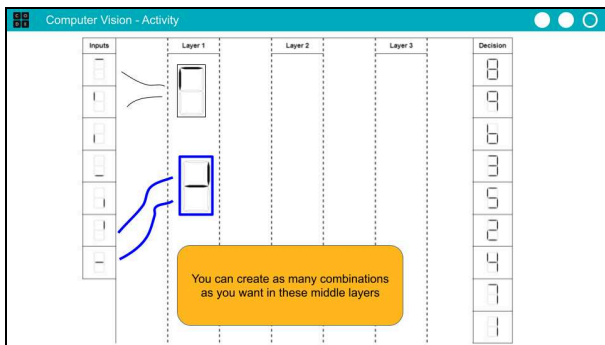
Say: The right side of this guide represents the decision we're trying to make. We need to build a path from the inputs on the left to these decisions on the right.



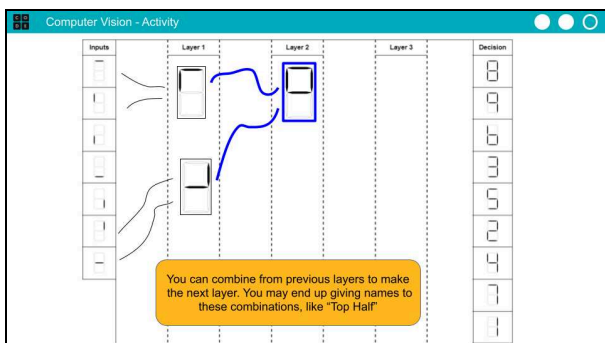
Say: This central area is where you can build your network, just like in the video



Say: You can combine at most 2 inputs from previous layers to create a new combination, like this "corner" combined from two segments



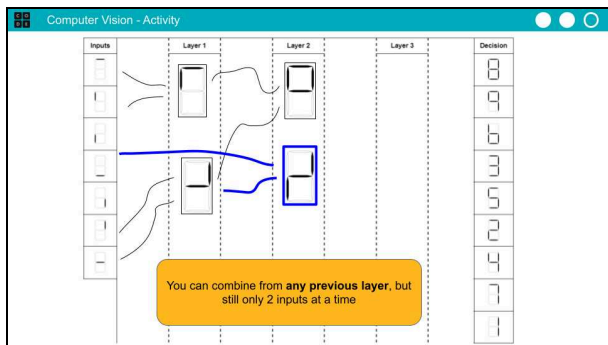
Say: You can make as many items in this column as you want! For the first layer, you may end up with a lot of combinations in this column



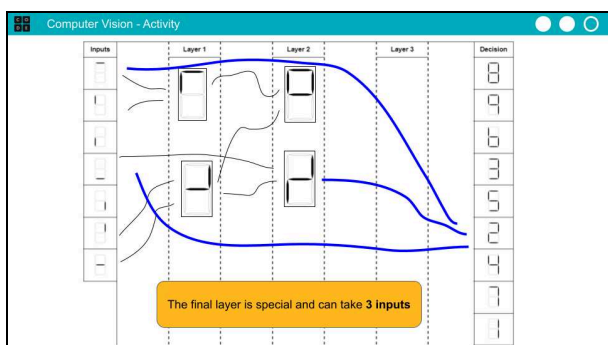
Say: You can combine from previous layers to move into the next layers, creating more complex shapes like this "Top Half"

Slides

Say



Say: You can combine from any previous layer, but you can still only use 2 inputs at a time



Say: The final layer is special - it can combine up to 3 inputs from any of the previous layers

Teaching Tip

Examples: Several examples are provided in this document: [\[Example\] Computer Vision Network](#). Consider reviewing these examples before teaching the lesson to see what kinds of networks students may create.

☒ **Do This:** Have students work in pairs to construct their network. The rules for the network are displayed on the slides.

☒ **Circulate:** Monitor students as they complete this task, observing the different strategies students attempt to "move through" the layers. Ensure partners are collaborating together. Look for groups that are creating unique or novel networks, especially when different from some of the examples in the presentation.

Teaching Tip

Why are they out of order? Students may ask why the numbers are out of order on the right side of the activity guide. They're ordered by "number of segments" - 8 has the most active segments, and 1 does not. When it comes time to make a decision, if multiple numbers are lit up (like 9 and 7 and 4 and 1), then the top number is always chosen. This helps reduce the complexity for students as they make their algorithms - otherwise, an 8 could cause havoc in their network!

Exemplar: A few example networks are provided as an Answer Keys. These can be used as reference, or to quickly inspire students who may be stuck. They can also be used in class discussions to emphasize that there are multiple ways to solve this task.

Regroup: Have students regroup

Teaching Tip

(Optional) Gallery Walk: Consider having students perform a gallery walk to visit other networks, and have students reflect on how the networks are similar or different from their own. This is an especially engaging activity if students have "Gone Big" and drafted their networks on whiteboards or poster paper.

Testing Our Algorithm

Remarks

Now that we have our algorithm, let's test it! We're going to do 2 examples. I'm going to show a slide that has certain inputs covered with post-it notes. This means the sensor detected that particular strand was lit up. Follow your network to determine which number it is!

 **Display:** Display the slide with the first example

(Click to expand)

Teaching Tip

Use the network, not your eyes: Students may comment that, once they have the inputs on the left-side, they already know what number it is - they can "figure it out just by looking at it". Remind students that the goal isn't for **us** to figure it out with our eyes and brain, but for a **computer** to figure it out with its sensors and algorithm. This can be useful for images that are incredibly large or complex (like satellite images of crowded markets) that we might not be able to interpret, or situations where we need to interpret images incredibly quickly which computers can do faster than we can.

You can also ask students to consider all of the steps that happened in nanoseconds once they saw these inputs - their brain needed to combine all of this information together to know what number it was. This network is doing the same thing, acting like the computer's "brain" to make a decision once it sees the inputs.

 **Circulate:** Monitor students as they trace the path in their network

 **Prompt:** *Are there any changes you want to make to your network? You have a minute to do so!*

 **Display:** Show the slide with the second example

(Click to expand)

Teaching Tip

Optional Extensions:

- Consider having students trade networks with another pair, then giving students a third example and seeing if they can trace a neighbor's network
- Consider inviting students to trace the solution in front of the class, explaining how their network works
- Consider giving students the number 8 as an input, which will activate every single space in their network, then prompt them to consider how they can make a decision when this happens
- Consider telling students that one of the segments is "burnt out", and then giving them a pattern that doesn't exactly match with any number (for example: an 8, but the bottom segment is missing, so it looks like an A). What number will their network identify it as?
- Consider asking students to "time themselves" tracing through the network, then asking students to notice if some networks are "more efficient" than others and take less time to complete.

Wrap up (5 minutes)

 **Prompt:** What do you imagine a network could look like that could identify the image below? What do the inputs look like? What do some of the middle layers look like?

(Click to expand. Image source: Wikimedia Commons)

 **Discussion Goal:** This is an open-ended prompt for students to apply the skills they practiced in the lesson to a new situation, but they still may struggle adapting to a new context - it's okay for students to struggle and "talk it out" as they work through this together. Students should identify the individual left and right flag positions as inputs, and the combination of the two as the output letter. The middle layer could look like two flags combined together - for example: left flag straight left + right flag straight right = R. This example network would be simpler than the class example because there are less components to build together.

 Content Corner 

Semaphores: This example comes from a historical way for ships or outposts to communicate over long distances, before radio or telephone were invented. For example, if two ships on the sea wanted to communicate, one person would wave flags according to this system and someone on another ship would record the words to piece together the message.

(Optional) Extension: Consider watching the [What a Neural Network Sees](#) video from the [Experiments with Google](#) page. The short video gives a live example of how image classification works as a neural network, where you can see the inputs and hidden layers.

After the Lesson

Teacher Survey

We'd love to learn more about the folks teaching these lessons and the classroom experience. Please let us know in this [How AI Works Teacher Survey](#).

Additional Lessons

If you'd like to teach additional lessons from the How AI Works video series, [click here to explore additional resources](#)

AI and Machine Learning Unit

If you'd like to dive even deeper into AI and Machine Learning, consider exploring our 5-week unit on AI and Machine Learning. [Click here to view the unit and learn more](#)



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Lesson 3: Neural Networks

45 minutes

Overview

This lesson centers around the [How AI Works: Neural Networks](#) video from the [How AI Works video series](#). Watch this video first before exploring the lesson plan.

Students learn how neural networks work. They first discuss an example of an experience that recommends things to you. They then use a widget that recommends videos based on one person. Students watch a video explaining neural networks. They use an updated widget to adjust the weights of each person. Finally, students discuss the need for diverse perspectives when creating recommendation systems.

This lesson can be taught on its own, or as part of a 7-lesson sequence on How AI Works - [click here to view all lessons in this sequence](#).

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- ▶ **AP** - Algorithms & Programming
- ▶ **CS** - Computing Systems
- ▶ **DA** - Data & Analysis
- ▶ **IC** - Impacts of Computing

Agenda

Before the Lesson

Warm Up (5 minutes)

Recommendations

Neural Networks (35 minutes)

WeTube (Beta)

WeTube 2.0

Wrap Up (5 minutes)

After the Lesson

Teacher Survey

Additional Lessons

AI and Machine Learning Unit

Objectives

Students will be able to:

- Explain the need for diverse perspectives when creating recommendation systems
- Use the weights of an artificial neuron to adjust the outputs of a recommendation system

Preparation

- Print copies of the WeTube Content Creators resource for each student
- Print copies of the Neural Network activity guide for each student

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- [Neural Networks](#) - Slides

▼ Make a Copy

For the students

- [How AI Works: How Neural Networks Work](#) - Video
- [Neural Networks](#) - Activity Guide

▼ Make a Copy

- [WeTube Content Creators](#) - Resource

▼ Make a Copy

Teaching Guide

Before the Lesson

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Supports for Teachers: You can learn more about bringing AI lessons to your classroom through our [AI 101 for Teachers series](#). AI 101 for Teachers is a free, foundational online professional learning series for any teacher and educator interested in the groundbreaking world of artificial intelligence (AI) and its transformative potential in education.

Warm Up (5 minutes)

Recommendations

 **Prompt:** Think of a time when you got a recommendation (for a video, a food, etc.) from either another person or a website or an app. Think of 1-2 examples to share.

 **Discussion Goal:** Encourage students to think of things that make recommendations, especially different apps or websites. For example, they may be recommended videos from Youtube or new songs on a music app. Some apps may ask you to pick preferences before using them to help figure out your preferences. Some websites give recommendations at the end of an experience, like a takeout menu or online grocery store recommending new items before checking out. Try to crowdsource a variety of examples before continuing to the next prompt

 **Prompt:** Do these recommendations tend to be accurate? Or do they feel random?

 **Discussion Goal:** Responses here will vary, and there's no expected answer. Instead, focus on the tension between getting recommendations that are accurate and how not every recommendation may feel like it matches each student's interests and personality.

Remarks

We get recommended things all the time, but how do these programs know what to recommend? How do they get better? Today we're going to look at one way computers create recommendations using something called neural networks.

Neural Networks (35 minutes)

WeTube (Beta)

 **Distribute:** Pass out the [WeTube Content Creators](#) resource.

 **Display:** Display the slide with information about WeTube, which is also on the top of the handout.

 **Model:** Show students the widget and demonstrate how to select a person and then view the video recommendations.

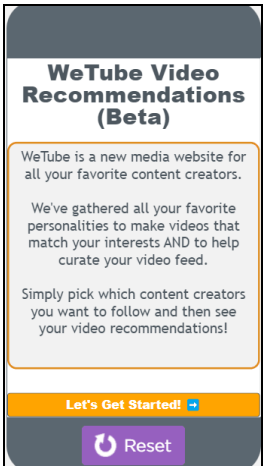
Teaching Tip

Real Or Fake? Students may have lots of questions about this business model and why it's a good idea to only go by recommendations. These may be valid questions if this were a real business, but it's not - assure students that this is just an excuse to focus on the role of recommendations in our world. However, it is true that the underlying concepts in this business and how recommendations work are similar to real-world recommendation systems, even if the company is fake.

Content Corner

Real-World Recommendation Systems: Spotify is one of many companies that uses a similar interface when users first use the app - it asks users to pick 4-5 different artists, and then curates songs based on those preferences.

The new Threads app from Meta is another example of a social media recommendation algorithm that works similarly to this program. In Threads, your feed is curated entirely by who you follow - there's no ability to search for phrases or further customize the feed yourself. Similarly, in this app, your feed is entirely determined by which artist you follow.

Widget	Say
	• Show students the introductory text, then click the "Let's Get Started" button to see the main screen • On the second screen, you can click on individuals to learn more about them. This is the same information that's in the level instructions and on the handout. • Once you've chosen a person, click the See Recommendations button at the bottom of the screen • This third screen shows the "video feed" that would appear based on this person's interests. The white boxes show video tiles, and the purple buttons can be used to scroll through multiple videos • Clicking the orange buttons on the left will filter the videos by certain categories. This is an easy way to show how preferences can change from one person to another. • After exploring some of the videos from the person you chose, click the "Select New Person" button at the bottom of the screen. Select a new person and repeat the process: show which videos have been selected with this new person

 **Code Studio:** Send students to the first level in Code Studio. As they work, have them consider the following prompt

 **Prompt:** Which content creator resonates with you the most and are most excited to follow? Which one resonates with you the least?

 1

WeTube Video Recommendations (Beta)

 **Circulate:** Walk around as students experiment with the app. Make sure they understand the interface - that after choosing a content creator, they see a list of video recommendations they can explore. Encourage students to explore different content creators and see what videos are recommended. Students should notice that the videos seem to match the personalities of each person with some variation. Once most students have looked at a few content creators, regroup the class.

Regroup: Have students regroup to discuss the following prompt

 **Prompt:** Ask the following questions in quick succession and having students respond non-verbally (for example, by raising their hands). The goal is to quickly move through these questions to motivate the next section of the lesson.

- Were there any videos that showed up for your person that you didn't think were super interesting or disagreed with?
- Were there any videos that showed up for another person that you wish had shown up with your favorite person?

Remarks

It's super unlikely that just subscribing to 1 person will give you the ideal recommendations - there may be some things here that you actually don't like, and you'd rather do something that a different user recommended instead. There's gotta be a way to combine these different perspectives to fine-tune the recommendations that we get. Let's look at how computer science and artificial intelligence can help with that!

 **Video:** Show the [How AI Works: How Neural Networks Work](#)

 Teaching Tip

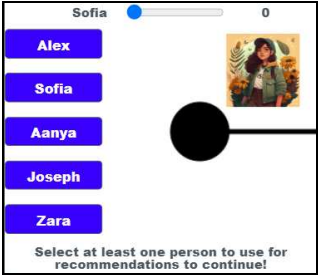
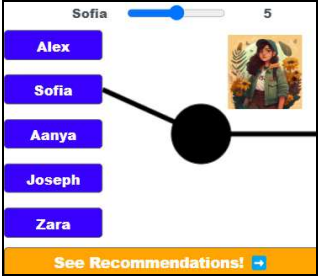
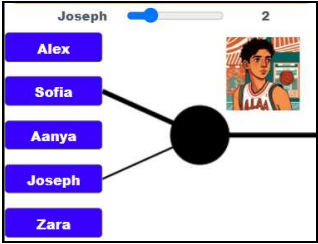
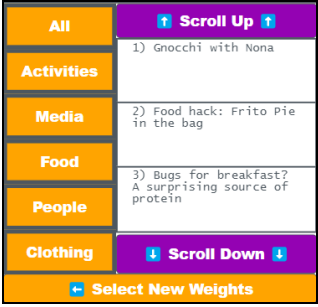
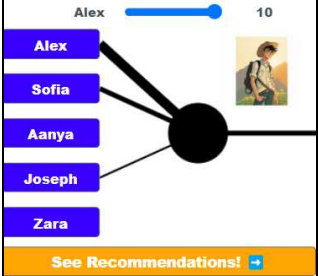
Videos are used throughout the curriculum to spark discussions, supplement key concepts with additional explanations and examples, and expose students to the various roles and backgrounds of individuals in computer science.

While interacting with the video, turn on closed captioning so students can also read along as they watch.

To encourage active engagement and reflection, use one or more of the strategies discussed in the [Guide to Curriculum Videos](#).

WeTube 2.0

 **Display:** Show the "WeTube Version 2.0" slide, which describes the updates to the widget. Click through the next slides to explain the differences in the widget.

Slide	Say
	 Display and Say: You can now combine recommendations from each person.
	 Display and Say: You can use the slider to adjust how much this person will influence your recommendations. Just like in the video, we're adding more weight to their recommendation
	 Display and Say: You can click on someone else and combine their recommendations. The thicker the line, the more they influence the recommendations.
	 Display and Say: Here are some of the recommendations we get with Sofia and Joseph. Both of their preferences are reflected here.
	 Display and Say: You can also click the Select New Weights button to go back and change your preferences. For example, we could decide to add Alex to our recommendation model.
	 Display and Say: When we do that, some new videos appear in our feed that were influenced by Alex's preferences.

 **Distribute:** Pass out the Neural Networks Activity Guide. Read through the overview at the top of the handout.

 **Code Studio:** Have students go to the next level in Code Studio.

 2

WeTube Video Recommendations (2.0)

Do This:

- Use the widget to create a custom recommendation system using each of the content creators.
- Combine their weights until you get a series of recommendations that you like and agree with.
- Test multiple combinations before settling on one that you like

Teaching Tip

How Is It Learning? This widget is similar to the example shown in the video and, behind the scenes, each person has given a rating to the different "videos" that get recommended. However, there is a key difference: in the video, the network customizes the weights by "learning" from how users rate the videos that are recommended to them, but in this widget users can manually adjust the weights to customize the network themselves. In this sense, the widget isn't really "learning" in the same way the video learns. However, it does create an equivalent type of recommendation system, and is similar to other recommendation systems that let users customize their interests themselves.

 **Circulate:** Monitor students as they complete this task. Students should be testing different combinations of weights and recommendations. Encourage students to share observations with their neighbors and really play with the combinations. Some probing questions could be:

- Are there any videos that appear for certain people?
- If you pick a particular video, is there a way to get it to the "front page" of your recommendations in your top 3?
- Are there videos that never seem to make it to the top 3?

Regroup: As students begin to settle on combinations that they like, regroup the class

 **Do This:** Jenna is a student who *loves* Boba Tea. There's a video called "Boba Tea Hacks: Unique flavor combinations". She's trying to find a combination that gets this video to the Top 3 videos. Can you help her?

 **Circulate:** Give students a few minutes to experiment here combining different content creators to get the video to appear in the top 3. Transparently: this video is designed to always be rated low by the current selection of content creators and never appear in the Top 3, so students won't actually be able to accomplish this task. This is intentional and designed to motivate the final part of the activity.

Remarks

It seemed like it was nearly impossible for Jenna to see her interest represented in her Top 3. Thinking about how Jenna must feel, it can be pretty disappointing to not be able to see your own interests represented in an app like this.

 **Prompt:** Have there been times where your interests or perspectives haven't been represented in an app? Why do you think that is?

Have students share with their neighbors

💡 **Discussion Goal:** Students may have a wide variety of experiences here. Some may be comical or inconsequential, like not having their favorite type of food recommended to them no matter how hard they try. Others may be more influential and consequential, like never having music or books or movies reflective of their identities or culture recommended back to them. Focus the discussion on why this might be, emphasizing answers that focus on the diversity of the data used for the recommendation algorithm. When this comes up, feel free to transition to the next part of the activity.

💡 Teaching Tip ▲

Reflecting on Social Media: Depending on student engagement in this discussion, this may be an opportunity to reflect on the impact of algorithmic recommendations and student participation in social media. Studies have shown that social media can **negatively impact teen mental health** (ie: seeing ads or recommendations that promote a negative self-image), and some students may describe themselves as addicted to social media.

Depending on your comfort with these topics, you could decide to facilitate a discussion with students. If so, consider using **Make Peace with Social Media** as a resource for both you and your students.

🎤 *Remarks*

Recommendation systems are only as good as the data they have to work with. In this example, none of our content creators liked the Boba Tea video enough to make it rise to the top, even though people like Jenna clearly want to watch it. Sometimes this causes people to leave the app entirely, but there's also another solution: adding more perspectives to the data!

☑ **Do This:** Flip the activity guide to the back side and read the overview as a class:

🎤 *Remarks*

You are now in charge of imagining a new content creator that WeTube will add to their recommendation system. This person could be based on an existing person (even yourself!), or could be totally imaginary. The key thing is that this person should represent new interests and perspectives not represented by the current group of people.

☑ **Do This:** Design your own content creator on the activity guide

🔄 **Circulate:** Check in with students as they complete this task, encouraging them to consider perspectives that aren't represented with the current group of content creators. This is also an opportunity to check in with and encourage students who may have been especially excited about the content of seeing themselves represented in a system like this, or may have felt their interests or culture historically excluded from existing recommendation systems.

💡 Teaching Tip ▲

Real-World Connections: Consider sharing with students the work of Michael Running Wolf, an Indigenous computer scientist who is working to add Native languages like Salish, Cheyenne, or Blackfoot to voice assistants. Part of this work includes creating the data that the assistants can use when generating their response, since that data was missing when these systems were first created. Consider sharing **this news article** with students or completing one of the activities from the **CS Education Week CS Heroes series**

Wrap Up (5 minutes)

🗣️ **Prompt:** Why is it important to have a diverse set of data when creating a recommendation system?

💡 **Discussion Goal:** This prompt is an opportunity to review and synthesize the last part of today's activity, where students contemplated how recommendation systems are dependent on the data used to create them. Encourage students to make connections between this prompt and some of the recommendation systems they brought up in the warm-up, considering how an expansive set of data can help make sure the system works for everyone instead of just a specific subset of people.

After the Lesson

Teacher Survey

We'd love to learn more about the folks teaching these lessons and the classroom experience. Please let us know in this [**How AI Works Teacher Survey**](#).

Additional Lessons

If you'd like to teach additional lessons from the How AI Works video series, [**click here to explore additional resources**](#)

AI and Machine Learning Unit

If you'd like to dive even deeper into AI and Machine Learning, consider exploring our 5-week unit on AI and Machine Learning. [**Click here to view the unit and learn more**](#)



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Lesson 4: Chatbots and Large Language Models (LLMs)

45 minutes

Overview

This lesson centers around the [How AI Works: Chatbots and Large Language Models](#) video from the [How AI Works video series](#). Watch this video first before exploring the lesson plan.

Large Language Models (LLMs) generate new text. The text LLMs generate looks like a human might have written it because large language models are built based on large bodies of text, such as Wikipedia. In this lesson, students learn what an LLM is and how it works, then use an LLM to co-create a text with AI. Finally, the class wraps up with a discussion about whether or not LLMs are intelligent or creative.

This lesson can be taught on its own, or as part of a 7-lesson sequence on How AI Works - [click here to view all lessons in this sequence](#).

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- ▶ **AP** - Algorithms & Programming
- ▶ **CS** - Computing Systems
- ▶ **IC** - Impacts of Computing

Agenda

Before the Lesson

Warm Up (5 minutes)

Activity (35 minutes)

Introduction to Large Language Models

Create With a Language Model

More about Large Language Models

Wrap Up (5 minutes)

After the Lesson

Teacher Survey

Additional Lessons

AI and Machine Learning Unit

(Optional) Extension Activity

Objectives

Students will be able to:

- Collaborate with AI to create a creative text document
- Describe that having a large amount of text to learn from allows computers to appear intelligent in written form
- Discuss positive and negative side effects of these models

Preparation

- Print copies of the activity guide for each student

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- [Generative LLM](#) - Slides
▼ Make a Copy
- [Written Voice Simulator User Guide](#) - Resource
▼ Make a Copy
- [\[Exemplar\] Written Voice Simulator Activity Guide](#) - Exemplar
▼ Make a Copy
- [\[Key\] Vocab And Video Guide](#) - Answer Key
▼ Make a Copy

For the students

- [\(Optional\) Eliza Activity Guide](#) - Activity Guide
▼ Make a Copy
- [How AI Works: Chatbots and Large Language Models \(Part 1\)](#) - Video
(Download)

- [How AI Works: Chatbots and Large Language Models \(Part 2\)](#) - Video ([Download](#))
- [Video and Vocabulary Guide](#) - Activity Guide [▼ Make a Copy](#)
- [Written Voice Simulator Guide](#) - Activity Guide [▼ Make a Copy](#)

Vocabulary

- **Large Language Model (LLM)** - a general purpose form of AI that is trained on a very, very large body of knowledge
- **Model** - a computer program designed to make a decision

Teaching Guide

Before the Lesson

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Warm Up (5 minutes)

 **Prompt:** Complete the following sentences with one word only:

- Thank ____.
- Please buy me a ____.
- Video games are ____.
- I enjoy playing outside in the Spring because it is ____.

 **Discuss:** Compare your answers with your neighbors. Where did you pick the same words? Where did you pick different ones?

💡 **Discussion Goal:** There are no right or wrong answers here; the idea is to collect a diversity of responses. This activity is meant to make students think about what words “probably” follow other words, and why. In this lesson, students will be learning about Large Language Models (LLMs), and how they guess likely next words based on context and what words have been close to each other in the past.

💡 Teaching Tip ▲

A **snowball discussion protocol** might work well here – give students 30 seconds to come up with their own answers, a minute to talk about them in pairs, then a minute for pairs to pair up with other pairs, until at 4 minutes you bring the whole class together.

🗣️ **Discuss:** Why did you pick these words?

💡 **Discussion Goal:** Students will likely have similar answers for the first sentence, but will diverge quickly with the other answers, such as which thing to buy. Emphasize how some prompts rely on the context of the sentence, like how students may have different opinions about video games, but they are more likely to give a positive answer to the last sentence because of the word 'enjoy'.

💡 Teaching Tip ▲

Optional Extension: Semantris Game: **Semantris** is a game developed by Google that emphasizes the same ideas in this warm-up: we naturally associate certain words with each other, which can be useful in a variety of situations. Semantris accomplishes this by generating a keyword, then the player tries to think of a word that's "close" to the generated keyword to score points. The game is engaging and fun. You could decide to play a few rounds as a class, or have students play for a bit on their own. If so, be careful of how much time you'd like to devote to this activity.

🎤 *Remarks*

This prompt is one example of how we can use our prior knowledge to guess what words come next in a sentence. We've also seen that those words will not always be the same. In today's lesson, we're going to look at how computers can write text based on a similar principle – learning from a lot of text that has been written before in order to guess what words might follow each other.

Activity (35 minutes)

💡 Teaching Tip ▲

Lesson Timing: This lesson is designed to fit within a 45 minute class period at a very brisk pace. As you look through some of the optional extensions or additional room for creativity and exploration, you may decide to extend this lesson over multiple class periods to fully explore all of the resources provided in this lesson.

Introduction to Large Language Models (10 minutes)

📄 **Distribute:** Pass out a copy of **Video and Vocabulary Guide** to each student

Remarks

We're going to watch a video introducing the big ideas for today. These ideas can be complex so we're going to pause the video a few times and take some notes as we watch. We're going to start even before the video starts by talking about what a **model** is.

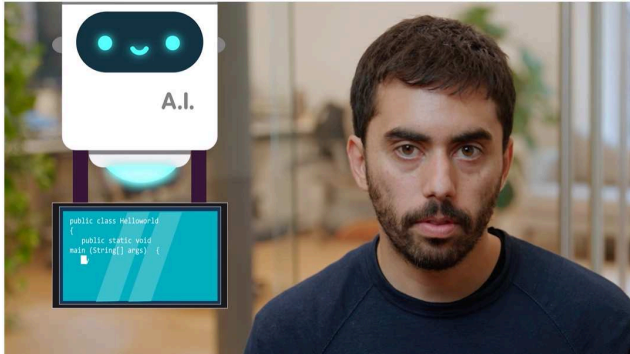
Vocabulary:

- **Model** - a computer program designed to make a decision

Teaching Tip

There are lots of ways to encourage students to retain vocabulary, and different methods will work well for different students. Encourage students to rephrase these definitions in their own words, draw pictures, or write sentences with these words. There is space on the handout for this.

Video: Show the [How AI Works: Chatbots and Large Language Models \(Part 1\)](#) video. At key points throughout (described below) we will be pausing the video to give students a chance to synthesize their understanding. They can keep track of their answers in the [Video and Vocabulary Guide](#)

Video	Teacher
<i>Pause video at 1:17 when they say "The possibilities seem endless"</i> 	Discuss: What is a large language model? Discussion Goal: Look for a quick answer to check for understanding of the video so far Students should explain that a large language model learns from a very large body of text (such as all of wikipedia) in order to generate new text.
<i>Let the video finish</i> 	Discuss: • How is the model picking the next letter? • What seems to be the problem with this method? Discussion Goal: Look for a quick check for understanding. Students should mention that the model is picking the next letter randomly, and they have probably noticed that these random picks of letters are not generating real words.

Create With a Language Model (15 minutes)

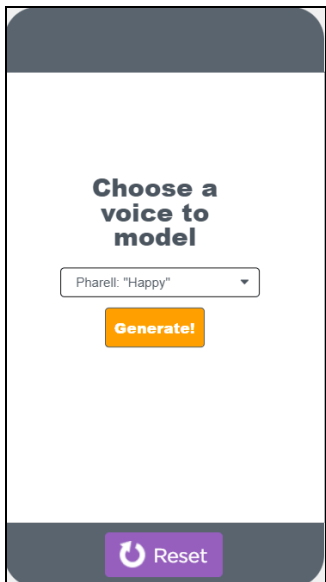
Remarks

The Shakespeare model in the video is guessing a possible next letter from a list based on what Shakespeare has said before. Let's do something similar, but with whole words, based on a written work of our choice.

 **Model:** Open up Code Studio to the Written Voice Simulator widget.

 1

Written Voice Simulator Widget

Widget	Modeling Notes
	• Demonstrate how to select a voice from the dropdown menu on the first screen. Press the Generate button to begin interacting with the widget • The widget starts with a random word, and then reveals the 5 most common words that appear <i>after</i> this word. Clicking one of these words will add it to the story and re-generate the buttons • Take suggestions from students on which words to click until a 10-12 word sentence is generated • Show how to return to the home screen by pressing the Change Voice button. Select a new voice and show how it can be added to the original story • Continue to take suggestions from students, then model how to delete the story so far. <i>Note: A full explanation of this widget is available in this Written Voice Simulator User Guide</i>

Teaching Tip

Citing our Sources: The app is a very small version of [Botnik](#), which uses a predictive text engine based on previous text from given tv shows, authors, video games, etc. This [video](#) provides more context around Botnik, starting around 1:20.

Remarks

This widget lets us generate text similar to what we saw in the video, and we can certainly use it to generate random or non-sensical phrases. Our goal is to try and make it generate sentences or phrases that are more cohesive or even creative, similar to poetry or song lyrics. We're going to use this handout to help guide our exploration.

 **Distribute:** Give each student a copy of the [Written Voice Simulator Guide](#).

 **Code Studio:** Have students log into Code Studio and begin interacting with the widget using the prompts from their activity guide.

Teaching Tip

After giving instructions, it can be helpful to check that your students understand what is expected of them – questions like “How long do we have to do this activity?” “What is expected at the end of it?” help to make sure the instructions are clear and the students are ready to get to work.

 **Circulate:** As the students write their stories, encourage them by asking for their thoughts and ideas, making sure they stay on task.

A link to the [Written Voice Simulator User Guide](#) resource is provided on the widget page. Students can access this for some of the advanced tasks on the worksheet that you may not have demo'd, like saving a story or loading a custom story.

There is a spot for students to share their stories with their neighbors and reflect on the similarities and differences. Make sure groups are doing this, or consider regrouping the class for a moment and ensuring all pairs get a chance to compare stories.

Regroup: Have students close out of the widget and re-focus as a whole class

 **Discuss:** When you and your neighbor compared your "sad" stories on the bottom of the first page: What did these stories have in common? How were they different? Why?

 **Discussion Goal:** Students have probably found that stories told by the same "voice" had more similar words, but choosing different words still meant different stories came out. This is to help students begin to have a framework around the Wrap Up Question, about whether LLMs are Creative and/or intelligent.



Content Corner



Language Model Games: The model that these programs build computes a "distance" between words – words that often follow each other, or are often used in place of each other, will have a shorter "distance." This process can be simulated with several activities:

- Google's [Semantris](#) is a game that visualizes what words Google's model thinks are closer to each other. You play by guessing a word that is "closest" to a target word in a list.
- [Semantle](#) is a Wordle-ish variant that allows players to guess a secret word by finding words that are slowly getting closer in distance.
- [Mind Meld](#) is an improv game that has two people say two random words at the same time, then try to find the word that is most common or "between" these two other words. They repeat this process - saying both words at the same time - with the goal of both people saying the same word at the same time. This means they've successfully "closed the distance" from their starting words by successively finding words they had in common.

All of these are optional and not part of the lesson plan, but can be interesting or fun "brain breaks" during other lessons or downtime in class.

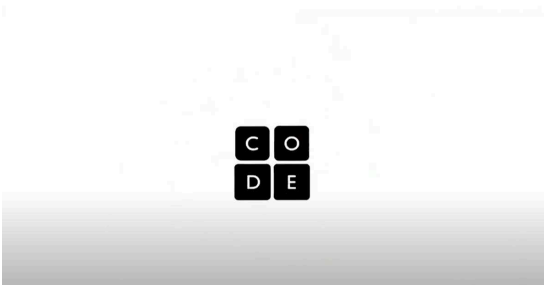
Digging Deeper: If you would like more information about the models behind Large Language Models, the YouTube channel [AI Coffee Break with Letitia](#) has a lot of great explanatory videos on this topic.

More about Large Language Models (10 minutes)

Remarks

This activity may have left you with more questions than answers. How can just combining words based on probability lead to chatbots that look and sound like real people? Let's watch more of the video to learn more!

 **Video:** Show the [How AI Works: Chatbots and Large Language Models \(Part 2\)](#) video. At key points throughout (described below) we will be pausing the video to give students a chance to synthesize their understanding. They can keep track of their answers in the [Video and Vocabulary Guide](#)

Video	Teacher
Pause video at 1:55 when they say "As you see, this works surprisingly well!" 	 Discuss: How does the system come up with better results?  Discussion Goal: Use this question as a quick check for understanding. Better results come from context - in this case, all of the words that came before it.
Pause video at 2:58 when they say "this system is still just using random probabilities to choose words" 	 Discuss: What are the three ways a large language model is different from the letter-by-letter Shakespeare example?  Discussion Goal: Use this as another quick check for understanding: 1. Rather than looking just at Shakespeare, it looks at everything it can find on the internet. 2. Instead of predicting letters, it predicts whole words or parts of words. 3. There is a lot of human tuning to look for biased or dangerous responses.
Let the video end 	 Discuss: Based on what you've seen so far, do you think a large language model is "intelligent"?  Discussion Goal: Students may share a variety of experiences, but based solely on the video and experimenting with the widget, it might be more accurate to say that LLMs generate text that sounds plausibly like a human might have written it, but may not actually be "intelligent"

 **Vocabulary:** Review the following vocabulary from the video. Allow students to check their answers from the activity guide.

- **Large Language Model (LLM):** a general purpose form of AI that is trained on a very, very large body of knowledge.
- **Context:** Additional information that helps a model make a decision

Assessment Opportunity

(Optional) Additional Prompt: If it seems like students are still somewhat unsure about large language models, consider facilitating the following discussion to help tease out additional key details:

 **Discuss:** How was the Written Voice Simulator also different from a Large Language Model?

 **Discussion Goal:**

- An LLM is usually learning from more words, but the Written Voice Simulator was only using a single source at a time (like the Shakespeare model)
- An LLM has context and can look at entire sentences and paragraphs. The Written Voice Simulator only knew one word of context when it gives us options.

- The way humans were involved is different. With LLMs, the bot is ultimately deciding which word to choose while humans can help influence the choices it has. In the Written Voice Simulator, it's reversed - the bot influences what choices the humans can make for what word comes next.

 **Discuss:** LLMs are learning from every word on the internet. What is a benefit of using the entire internet as input? What is a drawback?

 **Discussion Goal:** Learning from every word on the internet gives Large Language Models a ton of information to learn from, but not all of that information is factual, and not all the wording will be kind or appropriate.

Also encourage students to consider how people with access to writing on the internet tend to be more privileged voices, and therefore voices from more marginalized communities may not be as prevalent, which only further marginalizes these groups.

Wrap Up (5 minutes)

 **Prompt:** What's an example of something you've interacted with that probably uses an LLM? What things can you imagine using an LLM for in the future?

 Teaching Tip

This might be a good discussion to use the **Whip Around Protocol**.

 **Discussion Goal:** This discussion connects the video to things the students have experienced – they have probably seen predictive text in emails or cell phones, and have probably heard of ChatGPT, for instance. Students may imagine using this technology for anything that generates text, including writing code or news articles. If students are struggling to find answers, you can provide the examples of predictive text, and / or demonstrate.

After the Lesson

Teacher Survey

We'd love to learn more about the folks teaching these lessons and the classroom experience. Please let us know in this **How AI Works Teacher Survey**.

Additional Lessons

If you'd like to teach additional lessons from the How AI Works video series, **click here to explore additional resources**

AI and Machine Learning Unit

If you'd like to dive even deeper into AI and Machine Learning, consider exploring our 5-week unit on AI and Machine Learning. **Click here to view the unit and learn more**

(Optional) Extension Activity

Eliza: A Chatbot that is not an LLM (10 minutes)

Remarks

Most of the things that generate text today use a Large Language Model, but before we had the computing power to allow models to learn from the entire text of wikipedia, there were text-based AIs that were not based on Large Language Models.

Instead, these programs have rules for what sorts of text make sense at different times.

One of the most famous examples is Eliza, who was a computer program that was supposed to sound like a therapist that was invented in 1966. It was very impressive in its time. Let's explore what Eliza is and is not capable of.



(Optional) Eliza Widget

Option 1: Pull up an example Eliza displayed on the projector, and take responses from students for what to say to Eliza.

Option 2 (takes more time): Allow students to play with Eliza on their own computers for a few minutes. Follow along in the [\(Optional\) Eliza Activity Guide](#)

 **Discuss:** How did using Eliza feel different from using a Large Language Model? How does Eliza feel similar to using a Large Language Model?

 **Discussion Goal:** Students will probably notice that the responses are more limited, though they may also notice that it can still feel real sometimes.

Content Corner

Real-World Examples: Chatbots like this have been used over time for things like customer service, scheduling appointments, or even emotional comfort. An overview of various chatbots and the challenges involved are presented in this [TedEd Video on whether a computer can pass as a human](#).

When Eliza was first released, it needed warnings and disclaimers that it wasn't a real therapist and shouldn't be used for actual therapy. More recently, an [Eating Disorder hotline tried to replace it's human helpline with an automated chatbot](#), but then needed to shut down their program when the chatbot gave inappropriate advice.



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Lesson 5: Generative Images

45 minutes

Overview

This lesson centers around the [How AI Works: Creativity and Imagination?](#) video from the [How AI Works video series](#). Watch this video first before exploring the lesson plan.

Diffusion models generate images. Diffusion AI converts an image to noise, and trains an AI to reverse the process. In this lesson, students learn how AI can generate images, then explore a diffusion AI widget. Finally, the class wraps up with a discussion about whether or not these models are creative.

This lesson can be taught on its own, or as part of a 7-lesson sequence on How AI Works - [click here to view all lessons in this sequence](#).

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- ▶ **CS** - Computing Systems
- ▶ **IC** - Impacts of Computing

Agenda

Before the Lesson

Warm Up (5 minutes)

Activity (35 minutes)

Show the Video and discuss

Explore Diffusion

Wrap-up (5 minutes)

After the Lesson

Teacher Survey

Additional Lessons

AI and Machine Learning Unit

(Optional) GAN Activity

(Optional) GAN Activity

Objectives

Students will be able to:

- Describe that AI can generate fairly realistic looking images through a process known as diffusion – where a model learns how to draw details by reversing details
- Discuss whether or not these models are creative and who owns these images

Preparation

- Print copies of the activity guide for each student

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **Generative Images** - Slides
▼ Make a Copy
- **[Exemplar] Noisy Image Activity** - Exemplar
▼ Make a Copy
- **[Key] Generative Images Activity Guide** - Answer Key
▼ Make a Copy

For the students

- **(Optional) GAN Game Activity Guide** - Activity Guide
▼ Make a Copy
- **How AI Works: Creativity and Imagination** - Video ([Download](#))
- **Noisy Image Widget Activity Guide** - Activity Guide
▼ Make a Copy
- **User Guide: Noise Image Widget** - Resource
▼ Make a Copy

- [Video and Vocabulary Guide](#) -

Activity Guide [▼ Make a Copy](#)

Vocabulary

- **Diffusion AI** - converting an image to noise, and training an AI to reverse the process
- **GAN (Generative Adversarial Network)** - a way of using two competing AI models to generate something

Teaching Guide

Before the Lesson

Getting Started with Code.org: Consider watching our [Getting Started with Code.org](#) video series for an overview of how to navigate lesson plans, setup a classroom section, and other important features of the Code.org platform. Each video also has a support article if you'd prefer to read or print instructions - [click here to learn more](#).

Setup a Classroom Section: You can use a class section in Code.org to manage your students, view their progress, and assign specific curriculum - [click here to learn more](#).

If you are using a learning management system, there may be additional steps to sync your classes with Code.org:

- [Click here](#) for steps to setup your classes with Google Classroom
- [Click Here](#) for steps to setup your classes with Clever

Become a Verified Teacher: Lesson plans and levels have additional resources and answer keys for Verified Teachers, which is quick process that verifies your position at an educational institution. [Click here to complete a form](#) and you should have access to verified teacher resources in ~1 business day.

Supports for Teachers: You can learn more about bringing AI lessons to your classroom through our [AI 101 for Teachers series](#). AI 101 for Teachers is a free, foundational online professional learning series for any teacher and educator interested in the groundbreaking world of artificial intelligence (AI) and its transformative potential in education.

Warm Up (5 minutes)

 Prompt:

1. Turn to a partner
2. Count down from 5, and at the count of 1, both call out the name of an animal
3. For 1 minute, each draw a mix of those 2 animals
4. Compare your images -- what details are similar and what are different?
5. Share your images with another partnership -- can they guess what the prompt was?



Content Corner



Gratitude: The Code.org curriculum team is grateful to computer science teacher Terence Stone, who first introduced this idea with great enthusiasm and hilarity at a Code.org training several years ago.

(Click the images below to see examples of what students may create)

 **Discuss:** Debrief with the students what they found while comparing images. Feel free to take only 1-2 responses and not dwell on this as a class; students should have already discussed with each other

 **Discussion Goal:** The lesson will be about how AI creates new images based on details in words. This activity has students doing something similar, to get them warmed up for the lesson itself. They can see what people created with the same prompt that was different, and think about these things as they work through the lesson.

Remarks

You all just created animals that didn't exist before you drew them. Increasingly, AI can do this sort of thing as well! Does that mean it's creative? Through this lesson, we'll explore *how* AI builds these new images, so we can answer this question

Teaching Tip

Optional Extension - Generating Images: You could consider using a generative image tool like Stable Diffusion to generate some images to emphasize the point in the remarks. A free demo version is available here: <https://stablediffusionweb.com/#demo>. However, this website is not always reliable (sometimes the website is down) and the images generated are not always school appropriate from the given prompt - we recommend vetting this resource on your own before considering to use with students.

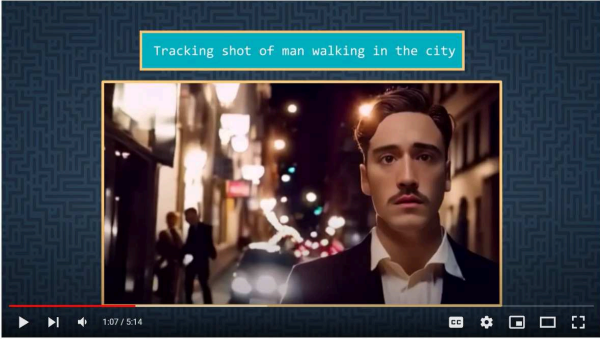
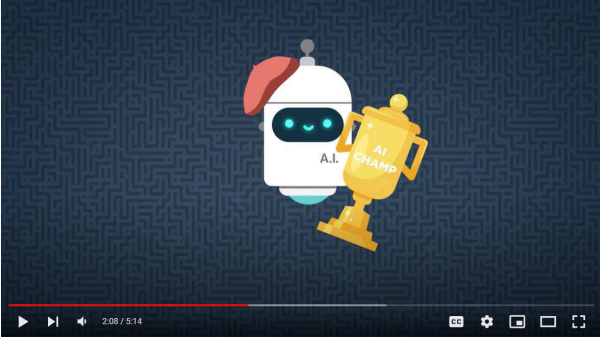
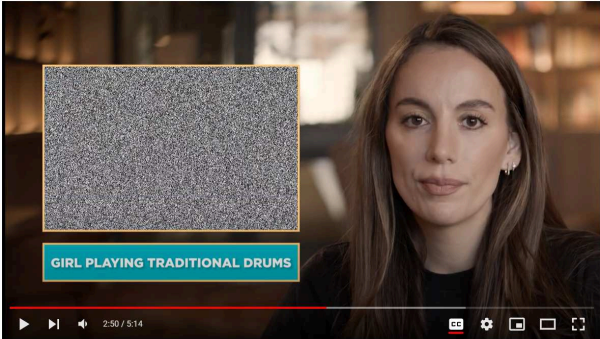
If you do decide to generate a few images, consider using a prompt similar to what students did in this warm up: "An animal that looks like a combination of a __ and a __ in a handdrawn cartoon style". You can take suggestions from the class to fill in the blanks. As students wonder "how is this happening?", you can transition to the rest of the activity which explains how AI like this works.

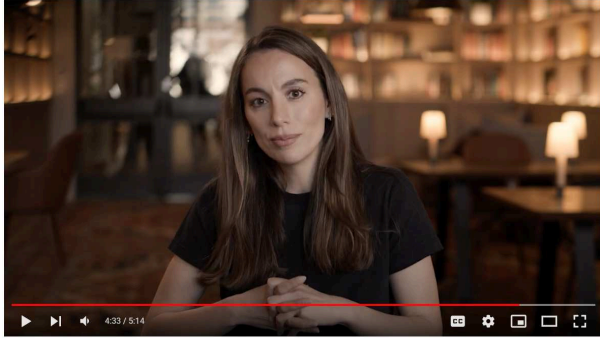
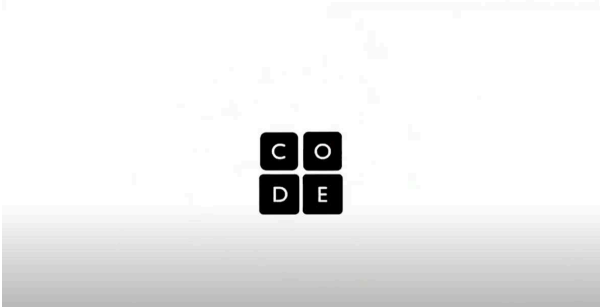
Activity (35 minutes)

Show the Video and discuss (20 minutes)

 **Distribute:** Hand out the **Video and Vocabulary Guide**

 **Video:** Show the **How AI Works: Creativity and Imagination** video. At key points throughout (described below) we will be pausing the video to give students a chance to synthesize their understanding on the activity guide.

Video	Teacher
Pause video at 1:10 when it says "And even create complete video sequences with just a few words of instruction." 	Prompt: What are some examples of things that AI can generate? Discussion Goal: Look for a quick answer to check for understanding of the video so far. Students may mention examples from the video such as building video sequences, but any place where humans provide AI with a prompt and it creates something will work. As a class, consider coming up with 1-2 examples not already stated in the video.
Pause video at 2:11 where it says "At this point, the generative AI has learned to generate real-looking imagery." 	Prompt: How did Generative Adversarial Networks make realistic-seeming images? Discussion Goal: This is a quick check for understanding. Students should know that a Generative Adversarial Network is made out of 2 networks -- one that generates images and another that says if it thinks the image was real or generated. Eventually, the generator is able to fool the discriminator and the image seem real.
Pause video at 2:53 when it says "...the image loses all details and becomes pure noise." 	Prompt: What is happening in each version of the image? Discussion Goal: This is a quick pause to allow students to synthesize what they are seeing. Students should note that the details in the images are getting slightly blurrier, but can still be seen. Eventually, the human eye can't see the details at all.
Pause video at 3:45 when it says "or to alter existing images." 	Prompt: How does the diffusion process train computers to build new images with specific details? Discussion Goal: This is a quick check for understanding. Students should know that diffusion models learn from slowly adding noise and then slowly subtracting noise from images, until the generator knows how to find details in noise.

Video	Teacher
Pause video at 4:36 when it says: "...so the legal questions here are not simple." 	 Discuss: Personal opinion: who do you think should "own" the images that AI creates?  Discussion Goal: There is no one answer to this question; the idea is for students to truly think about the implications of image generation in AI. Depending on your preferences and class timing, this can be an involved discussion that takes several minutes. See the Teacher tip for additional ideas for how to facilitate this discussion.
Continue watching the video until it ends 	 Prompt: What can you imagine creating with a diffusion model??  Discussion Goal: Get students excited about what diffusion models can do.

 **Vocabulary:** Display the vocabulary from the video so students can check their vocabulary notetaker.

- **GAN (Generative Adversarial Network)** - a way of using two competing AI models to generate something.
- **Diffusion AI** - converting an image to noise, and training an AI to reverse the process.

Teaching Tip

Optional Activity: The focus on this lesson should be on the diffusion model, not the GAN. But, if you believe you will have extra time and would like to further understand this version of image generation, you can run an **optional 20 minute GAN activity** described near the end of this lesson plan.

Why Diffusion? Students may wonder where the term "diffusion" comes from. If so, it may help to show a video of a physical diffusion process, such as food coloring dispersing in water (**such as in this video**). If you show this video or a demonstration, point out that the food coloring is somewhat randomly slowly distributing itself throughout the water. This is what will happen with noise being added to images in diffusion models.

Who Owns the Image? For this discussion, some possible answers are:

- the people who designed the images the AI learned on
- The programmer who created the tool
- The person who wrote the prompt
- none of the above

If you linger on this question during the video, consider using the **Nominal Group Technique** and circulate, listening for students who are saying particularly interesting things to bring up with the whole group.

Diving Deeper: Questions of ownership, copyright, and labor are hotly debated as generative AI continues to expand. Many generative AI systems used images from living artists and can generate art in a similar style, even though their images were used without their consent and without payment. If students are interested in this ethical question, consider discussing this article with students: **Artists Are Suing Artificial Intelligence Companies and...**

Explore Diffusion (15 minutes)

Remarks

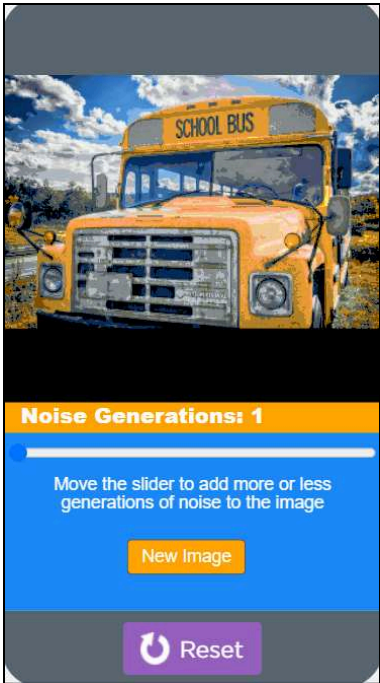
Today we're going to look closer at the diffusion process that is used by image-generating AI. This involves creating noise in an image. For this activity, we'll explore what you can see in an image with different levels of noise. Then, we'll guess other people's images based on what we can see at different levels of noise, just like an AI model does when it "generates" an image.

 **Distribute:** Hand out the [Noisy Image Widget Activity Guide](#)

 **Model:** Show how to use the Noisy Image Widget. Have the class fill out Part 1 of the activity guide as you demonstrate, asking them to describe what they can still see as you add more noise to the image.

 1

Noisy Image Widget

Widget	Modeling Notes
	• Demonstrate how to use the slider to add noise to an image. Point out the scale from 1 to 3000 for noise generation. • Show noise at different levels as shown in the activity guide, prompting the class to write what they can "still recognize" even with so much noise • Show how to add their own image via a URL. Consider having a URL ready, such as a school logo or class picture • Show how to repeat the noise generation process with a new image Note: A full explanation of this widget is available in this User Guide: Noisy Image Widget

 **Code Studio:** Have students log into code studio and interact with the Noisy Image Widget, completing Parts 2-4 on the activity guide.

Teaching Tip

Patience is a Virtue: The noisy images can take a short while to load here (usually less than a minute, but nothing is happening for that time). We recommend you warn students this is the case, and give them something to be focusing on, for example, choosing the next image / numbers, sketching / describing the image on the back of their worksheets, or solving side puzzles for extra credit.

 **Circulate:** Help student stay on task, guessing things about their images. Monitor them as they complete the activity guide and trade images with a partner. Make sure they are choosing school-appropriate images to share with neighbors.

Regroup: Have students regroup for a synthesis discussion

 **Discuss:** What did you notice during this activity?

Follow-up (if necessary): Were you able to see details in the noisy images? Did you see more details once you knew the original image?

 **Discussion Goal:** This discussion helps students synthesize what they were learning, and connect it back to the video on diffusion models finding details.

Wrap-up (5 minutes)

 **Discuss:** Are diffusion models *creative*? Why or why not?

 **Discussion Goal:** Students may have very different answers to this question. Some may think diffusion models are creative because they create things that have never been made before from prompts. Others will note that it's only re-creating details it has seen before in new configurations

After the Lesson

Teacher Survey

We'd love to learn more about the folks teaching these lessons and the classroom experience. Please let us know in this [How AI Works Teacher Survey](#).

Additional Lessons

If you'd like to teach additional lessons from the How AI Works video series, [click here to explore additional resources](#)

AI and Machine Learning Unit

If you'd like to dive even deeper into AI and Machine Learning, consider exploring our 5-week unit on AI and Machine Learning. [Click here to view the unit and learn more](#)

(Optional) GAN Activity

(Optional) GAN Activity (20 minutes)

 **Model:** Pass out the **(Optional) GAN Game Activity Guide**. Show students how to use the widget by playing one round of human guessing vs computer guessing. Feel free to use the slides at the end of the slide deck as a guide. Be sure to be clear to students that they have no information initially, and they do not have to over-think initial guesses. They'll get things wrong and that's ok! In fact, it's part of the model of the activity -- GANs are wildly guessing too until they have more information.

Exemplar: You can find an exemplar of this activity guide here: [GAN Game Activity Guide Exemplar](#)

 **Code Studio:** Students log into Code studio to explore a game where they act like generators.

 2

(Optional) GAN Guess Widget

 Teaching Tip

The rule that has been hard-coded into this activity is "must contain one red circle." Some students may figure this out before others. If this happens:

- You can have students who have completed the task to help ask other students leading questions to help them figure out the rule.
- You can have groups of completed students come up with their own password rules and figure out each other's in groups. This may be useful fun for all students, if you have extra time.

 **Circulate:** Help student stay on task, coming up with guesses with them. (Eventually Students should figure out that the rule is a password must have a red circle -- help students who are struggling figure this out by procedurally trying different shapes and colors with them.

Regroup: Have students regroup for a synthesis discussion to help them further cement their understanding of generators and discriminators by connecting it to what they did in the activity.

 **Prompt:** When were you acting like a generator in this activity?

 **Discussion Goal:** When students generated random passwords -- even when they weren't accepted -- they were acting like a generator. Generators generate random images that may or may not be accepted as real.

 **Prompt:** When were you acting like a discriminator in this activity?

 Teaching Tip

Common Misconception: Students will frequently assume that the computer telling them if their guess was correct or not was the computer acting as a discriminator. This is not true. Discriminators don't necessarily have the truth -- they are, like the students guessing if the computer's password will follow the rule or not, also making guesses. The computer providing the "FOLLOWS" or "NOT" feedback is acting more like the programmer that feeds the Discriminator labeled data to learn from. More information on labeled data exists in the machine learning video.

 **Discussion Goal:** Students acted like a discriminator when they guessed if the computer's generated password followed the rule or not. They didn't initially know, but got better as they got more data.



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Lesson 6: Algorithmic Bias

45 minutes

Overview

This lesson centers around the [How AI Works: Equal Access and Algorithmic Bias](#) video from the [How AI Works video series](#). Watch this video first before exploring the lesson plan.

In this lesson, students will practice cropping images to uncover the bias underlying the Twitter cropping algorithm. Then, students will read and watch a video about the discovery of this biased algorithm. Finally, students will discuss ways to recognize and reduce bias along with analyzing Twitter's response to the allegations of bias in their cropping algorithm.

This lesson can be taught on its own, or as part of a 7-lesson sequence on How AI Works - [click here to view all lessons in this sequence](#).

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- IC - Impacts of Computing

Agenda

Before the Lesson

Warm Up (5 minutes)

Activity (35 minutes)

Cropping Widget

Twitter Cropping Algorithm

Wrap Up (5 minutes)

After the Lesson

Teacher Survey

Additional Lessons

AI and Machine Learning Unit

Objectives

Students will be able to:

- Reason about which types of tasks should not be completed by an algorithm

Preparation

- Complete the cropping widget activity yourself so you can see what your students will experience with the widget will be and what questions or issues they may bring up.
- Watch the parts of the video [“Are We Automating Racism?”](#) referenced in the lesson plan.
- Read the two articles from the links section below.

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- [Algorithmic Bias](#) - Slides
▼ Make a Copy
- [Sharing learnings about our image cropping algorithm \(Twitter\)](#) - Resource

For the students

- [How AI Works - Societal Impact](#) - Video
- [Twitter says its image-cropping algorithm was biased, so it's ditching it \(CNN\)](#) - Resource ([Download](#))

Teaching Guide

Before the Lesson

Getting Started with Code.org: Consider watching our [Getting Started with Code.org](#) video series for an overview of how to navigate lesson plans, setup a classroom section, and other important features of the Code.org platform. Each video also has a support article if you'd prefer to read or print instructions - [click here to learn more](#).

Setup a Classroom Section: You can use a class section in Code.org to manage your students, view their progress, and assign specific curriculum - [click here to learn more](#).

If you are using a learning management system, there may be additional steps to sync your classes with Code.org:

- [Click here](#) for steps to setup your classes with Google Classroom
- [Click Here](#) for steps to setup your classes with Clever

Become a Verified Teacher: Lesson plans and levels have additional resources and answer keys for Verified Teachers, which is quick process that verifies your position at an educational institution. [Click here to complete a form](#) and you should have access to verified teacher resources in ~1 business day.

Supports for Teachers: You can learn more about bringing AI lessons to your classroom through our [AI 101 for Teachers series](#). AI 101 for Teachers is a free, foundational online professional learning series for any teacher and educator interested in the groundbreaking world of artificial intelligence (AI) and its transformative potential in education.

Warm Up (5 minutes)

 **Discuss:** *Think about a time when you shared a picture with your friends or on social media. Have you ever had to crop a photo you shared? What did you crop out and why?*

 **Discussion Goal:** Use this discussion to show that users usually have a reason for sharing photos and the focus depends on that reason. Students will most likely bring up a wide range of examples that can be used to highlight how there isn't a one size fits all approach to sharing photos and cropping them.

Remarks

Today we are going to try our own hands at cropping and then closely examine a real-world situation where training a model to crop photos leads to algorithmic bias.

Activity (35 minutes)

Cropping Widget (10 minutes)

 **Code Studio:** Have students log into Code Studio and begin interacting with the Cropping Widget. Each student should interact with the widget on their own, but they will be discussing with a partner.

 1

Cropping Widget

☒ **Do This:** Tell students to use the widget to crop the images as if they were uploading them to a social media account.

 **Circulate:** Monitor students as they complete this task. Ask probing questions as you notice students pause on certain images, asking what they're thinking or what they notice in the image, or what causes them to pause or be unsure.

 Teaching Tip

Intentionally Mischievous: The first few images are pretty innocuous with how students may decide to crop images, but as the images continue, the decisions become harder and harder. Students may decide to prioritize faces or text or color or foreground vs background when cropping. But eventually, images will deliberately pit student patterns against each other and force some uncomfortable decisions when deciding what information is important (ie: belongs in the frame), and what information isn't (ie: can be cropped out).

Lean Into Discomfort: Students may feel unsure how to crop certain images and may even feel slightly uncomfortable with a decision. This is okay and expected - the goal is to surface that even something as seemingly simple as "cropping a photo" can contain complex decisions. Emphasize that it's the *decision itself* - that the system is forcing them to decide in the first place - that creates this uncertainty, and not whatever they happen to decide with each individual picture.

Regroup: As you notice students reaching the end of the activity, pause the activity and regroup.

 **Discuss:** Which images were challenging for you to decide how to crop? Which images did you and your partner crop the same? Which images did you and your partner crop differently?

 **Discussion Goal:** Use this discussion to highlight the different experiences and perspectives everyone brings to deciding how to crop an image. Direct students to consider not only what they decided to keep in the picture, but also what they decided to crop out of the picture. What happens when we crop something out of a picture? It never gets seen by the audience. Look back at the photos you cropped. What or who got left out? You may choose to guide the class toward describing general characteristics of images that were challenging to crop.

 Remarks

Now imagine we recorded how everyone in this room cropped the images from this widget. We could use that information to train a model. This model could be used to crop other photos being uploaded to the social media site.

 **Discuss:** What patterns do you think might emerge from the data we collectively contribute to this model?

 **Discussion Goal:** Help students generalize some of the patterns or "rules" they may have been using when cropping, such as:

- emphasizing faces
- emphasizing text
- emphasizing things in light vs dark
- emphasizing things in the foreground instead of background

Depending on your level of comfort with the class, you could probe implicit biases with how the class may have been cropping, such as:

- did the class tend to prefer the left-side of the screen or the right-side of the screen?
- did the class tend to prefer younger faces, older faces, or faces their age?
- did the class tend to prefer men or women or crops without people at all?

Twitter Cropping Algorithm (25 minutes)

 Remarks

It turns out that the issue of how to crop an image is something social media platforms have been working on for some time. We are going to watch a video that shows the issues that arose when Twitter used machine learning to train an algorithm to do this cropping.

🎥 **Video Part 1:** Watch the video [“Are We Automating Racism?”](#) from 0:00-4:05

🎥 **Video Part 2:** Watch the video [“Are We Automating Racism?”](#) from 9:30-13:46

📄 **Display:** Show the next slide, which shows a definition that to use in the following remark.

🎤 *Remarks*

In the video, the primary reporter Jos asks her two colleagues, “So do you think this machine is racist?” We might all have different definitions of the term racism, but for the purposes of this discussion we are going to use this definition of racism - prejudice, discrimination, or antagonism directed against a person or people on the basis of their membership in a particular racial or ethnic group, typically one that is a minority or marginalized.

💬 **Discuss:** Using this definition of racism, what aspect(s) of racism did the Twitter cropping algorithm do: prejudice, discrimination, or antagonism?

💡 **Discussion Goal:** Students should gravitate towards prejudice and discrimination, since one group is being given different treatment purely based on the color of their skin according to the twitter algorithm. Use this discussion to highlight the evidence brought up in the video both from the reporters and the tweets they mention from other users showing how the algorithm was biased or prejudiced towards lighter faces compared with darker faces.

💡 Teaching Tip

Lived Experiences: Students may bring in other examples from their own lives where technology has appeared biased or prejudiced against them based on their race or skin color, especially involving sensors and image recognition. Facial recognition in particular has a history of having racial biases against dark-skinned people, which may have impacted students in your classroom. Consider how you may want to validate the lived experiences of the students in your room while continuing to guide the conversation towards the solution-focused section of the lesson, where Twitter ultimately implemented a solution for this particular algorithm.

Looking to the Future: This activity puts students in a position to see how innocuous problems (like cropping photos) can lead to unintended consequences, and students may be uncomfortable reflecting on their experience with the widget and the consequences of their decisions. Ensure students that these types of decisions are everywhere, beyond just this image and cropping algorithm issue, and it's how we respond that matters (as we'll see shortly with how Twitter responded). Remind students that they are future leaders and being aware of how bias creeps into decisions and into our data will make them better equipped to solve problems like these when they see them in the future.

💬 **Discuss:** How was the Twitter cropping algorithm trained? According to the video, where is a potential source of bias when training similar cropping algorithms?

💡 **Discussion Goal:** The most important thing for students to come away from this discussion with is the fact that the machine is not "racist", but it is biased towards lighter faces because of the datasets used to train it. Use this discussion to ensure students understand how the model used the saliency datasets discussed in the video and how some of those datasets were trained on photo libraries containing images that do not represent the population as a whole.

💬 **Prompt:** If you were the CEO of Twitter and found evidence of this bias in your cropping algorithm, how would you respond? What steps would you take?

You may choose to have students share their responses if there is time, otherwise move to reading the article.

🎤 *Remarks*

Now that you've had time to consider how you might respond, we are going to read about how this actually played out and Twitter's response when presented with evidence of bias in their cropping algorithm.

☑ **Do This:** Have students read the article [**"Twitter says its image-cropping algorithm was biased, so it's ditching it \(CNN\)"**](#). After they are finished reading they should mark up the text with the following:

- **Highlight / Underline:** Any information in this article that you want to know more about
- **At The End:** Write a 10-word summary of the article

💡 Teaching Tip

To Print or Not To Print? This lesson is written assuming that you have printed out the article and have it physically available for students to write on, even though it is also possible to have students interact with this text digitally. If students read the article digitally, it is most important that they still follow the active reading strategies outlined in this lesson - highlighting the text, writing in the margins, and summarizing. This may require some additional time & instruction to teach students your preferred tools of digital annotation, and may require some additional adjustments to some of the later annotation strategies in this lesson.

Wrap Up (5 minutes)

💬 **Discuss:** Have students discuss their reactions to the quotes pulled out from Rumman Chowdry, a software engineering director for Twitter's machine learning ethics, transparency, and accountability team.

- Do you think Twitter's response was appropriate?
- What are the risks and potential harms of systems, such as social media platforms like Twitter, becoming too dependent on machine learning?

🎤 Remarks

The story of Twitter and its photo cropping algorithm shows how something that's built with one intention, in this case to better highlight photos, can lead to unintended effects, in this case, bias of white faces over black faces, which can also have harmful impacts on society or culture. And, this case provides an example of a software company that has taken actionable steps to address algorithmic bias by establishing this team (machine learning ethics, transparency, and accountability team) lead by Rumman Chowdry.

💡 Teaching Tip

Diving Deeper - Rumman Chowdhury: Consider exploring the [**CSEdWeek CS Heroes resources on Rumman Chowdhury**](#) for additional activities, posters, and resources about Rumman, AI Ethics, and her work on bias in machine learning models.

🎥 **(Optional) Video:** Watch the video [**How AI Works - Societal Impact**](#) for an extended look at AI's impact on society and concerns about bias beyond just this example.

After the Lesson

Teacher Survey

We'd love to learn more about the folks teaching these lessons and the classroom experience. Please let us know in this [**How AI Works Teacher Survey**](#).

Additional Lessons

If you'd like to teach additional lessons from the How AI Works video series, [click here to explore additional resources](#)

AI and Machine Learning Unit

If you'd like to dive even deeper into AI and Machine Learning, consider exploring our 5-week unit on AI and Machine Learning. [Click here to view the unit and learn more](#)



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Lesson 7: Our AI Code of Ethics

45 minutes

Overview

This lesson centers around the [How AI Works: Privacy and the Future of Work](#) video from the [How AI Works video series](#). Watch this video first before exploring the lesson plan.

In small groups, students conduct research using articles and videos that expose ethical pitfalls in an Artificial Intelligence (AI) area of their choice. Afterward, there are options for how students synthesize what they've learned to make recommendations for how AI can be used in society: creating an AI Code of Ethics, or creating an AI Manifesto, or designing an "Imagine your World" artistic representation.

This lesson can be taught on its own, or as part of a 7-lesson sequence on How AI Works - [click here to view all lessons in this sequence](#).

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- IC - Impacts of Computing

Common Core English Language Arts Standards

- RI - Reading Informational

Agenda

Before the Lesson

[Preparing to Teach a Code.org Lesson](#)

[Preparing to Teach a Lesson on AI Impact & Bias](#)

Warm Up (5 minutes)

Activity (35 minutes)

[Introduction to Artificial Intelligence](#)

[Student Research & Reflection](#)

Option 1: AI Code of Ethics

Option 2: Create a Manifesto

Option 3: Imagine Your World Art Display

Wrap Up (5 minutes)

After the Lesson

[Teacher Survey](#)

Objectives

Students will be able to:

- Define artificial intelligence (AI) in their own words, using technologies they encounter in their daily lives as examples.
- Describe at least one example of an ethical issue pertaining to AI, along with its impact on society.

Preparation

- Review the slide deck and student resources
- Review the research topics and decide if you would like to remove / add any topics
- Decide which final activity to complete with the class (there are 3 options)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- [AI Code of Ethics](#) - Slides
▼ Make a Copy
- [LEADERS Framework](#) - Resource
▼ Make a Copy

For the students

- [AI Ethics Research Areas](#) - Activity Guide
▼ Make a Copy
- [Create a Manifesto](#) - Activity Guide
▼ Make a Copy
- [Ethics and AI: Privacy & the Future of Work](#) - Video

Additional Lessons

AI and Machine Learning Unit

- [Imagine Your World](#) - Activity Guide
▼ Make a Copy
- [Research Reflection Guide](#) - Activity Guide
▼ Make a Copy
- [\[Template\] AI Code of Ethics](#) - Slides
▼ Make a Copy

Vocabulary

- **Artificial Intelligence** - A technology that mimics human intelligence, performing tasks such as understanding language, recognizing patterns, and making decisions.

Teaching Guide

Before the Lesson

Preparing to Teach a Code.org Lesson

Getting Started with Code.org: Consider watching our [Getting Started with Code.org](#) video series for an overview of how to navigate lesson plans, setup a classroom section, and other important features of the Code.org platform. Each video also has a support article if you'd prefer to read or print instructions - [click here to learn more](#).

View the Resources: This lesson plan comes with a slide deck and additional resources, found in the Links section of the lesson plan. Each bolded line in the lesson plan usually corresponds to a slide in the slide deck. It can be helpful to have the slides and student resources open while reading the lesson plan, as there are visuals and additional information not always represented in the lesson plan.

Supports for Teachers: You can learn more about bringing AI lessons to your classroom through our [AI 101 for Teachers series](#). AI 101 for Teachers is a free, foundational online professional learning series for any teacher and educator interested in the groundbreaking world of artificial intelligence (AI) and its transformative potential in education.

Preparing to Teach a Lesson on AI Impact & Bias

Checking Resources: Students will be researching different topics related to Artificial Intelligence. Research areas are included in this handout: [[airesearchareas/ai-ethics/2023](#)]. We haven't provided these directly to students because sometimes these sites are blocked by district filtering programs, or the latest news-worthy innovation may not be middle-school appropriate. **We strongly suggest you check each site for appropriateness of content and to make sure it will load on classroom computers before sharing it with students.** You are also welcome to curate your own news sources based on current events or issues in your state or community.

LEADERS Framework: This lesson allows students to explore the societal impacts of artificial intelligence on society, both the intentional solutions that can be beneficial to society and the unintended consequences that may be harmful to society. Issues of bias in artificial intelligence may come up in research material and class discussion.

To help prepare for these moments, we recommend reviewing the [[r leaders_framework/ai-ethics/2023](#)] for facilitating lessons and discussions involving the societal effects of AI, including bias and ethical considerations. Not all of the tenets are applicable to every lesson, but many of the teaching tips and discussion goals in this lesson follow the principles outlined in this framework.

Warm Up (5 minutes)

 **Display:** On a whiteboard, draw a T-chart that says "Helpful" and "Harmful".

 **Prompt:** The Internet has been one of the greatest technological innovations in the last century. What are ways it has been helpful towards society? What are ways it has been harmful?

Teaching Tip

Differentiating the Topic: You could decide to change this prompt to use a different technology that is more relevant to your students and their interests, as long as there are clear examples of how it has been helpful or harmful in society. For example, you could ask the class for examples of a technology that's had a big impact on their lives (such as social media, maps, voice assistants, or self-driving cars) and then continue the warm-up using the class idea as the prompt.

You could also change the topic entirely to a relevant issue in your community - for example, if a new mining quarry might be opening near a community, this could be used as a topic to discuss ways it can be helpful or harmful (ie: boosts the economy, but may decrease the air quality in the community).

 **Discussion Goal:** Have students brainstorm individually, then share with a neighbor before regrouping. Call on students to share, keeping track of responses on the T-chart.

Students should be able to pull from news sources or lived experiences to populate the chart. If students struggle generating ideas, continue to the next slide which includes a few examples.

Teaching Tip

Context Matters: What helps one student may harm another, and vice versa. If students have conflicting ideas around "helpful" and "harmful", try not to open for direct debate, as this may result in the majority outweighing a vulnerable student. Instead, encourage students to think about future technologies, research, or policies that could protect people from being harmed. This will help segue into later activities encouraging students to be a part of the solution!

 **Display:** Show the slide with various news articles about the impact of the Internet on society. If you changed the prompt to a different technology, also update these slides.

Remarks

Even though the Internet is a grand technological innovation, its legacy is complicated - in some cases, it's been incredibly helpful and beneficial to society. And in others, it's been harmful and detrimental to society. Technology can have a wide-range of impacts, sometimes beyond what was intended. Today, we're going to example another piece of technology and explore the impacts its having on society: Artificial Intelligence.

Activity (35 minutes)

Introduction to Artificial Intelligence (5 minutes)

Teaching Tip

Skip to the Research: This section starts with a brief introduction of artificial intelligence. If you have already introduced artificial intelligence to students in previous lessons, consider skipping this section.

Connecting to your Subject: In order to expand on the definition of AI, you may want to bring up other examples to help students think about AI in their daily lives. You can even specifically discuss AI and your subject area. For example, English teachers can discuss AI-written essays. Math teachers can discuss how AI assists computation. Economics teachers can discuss how AI is used for financial prediction. History teachers can discuss the past decade of innovation in AI and how it has impacted society. You can simply think about the apps you use on a daily basis, as these are all opportunities to discuss AI.

 **Vocabulary:** Show students the definition of artificial intelligence.

- **Artificial Intelligence:** A technology that mimics human intelligence, performing tasks such as understanding language, recognizing patterns, and making decisions.

 **Prompt:** What are some examples of artificial intelligence you are already familiar with?

 **Discussion Goal:** Responses may include examples from movies and television, such as robots like WALL-E or R2D2. They may also include personal examples, like Siri on a phone or TikTok filters that can detect facial features. Try to surface examples that may not be immediately apparent but still govern ways we interact with the world, such as recommendation algorithms on social media, facial recognition, or times when a computer makes a decision.

Remarks

Today we're going to explore several specific areas of artificial intelligence. Our goal is to uncover how these technologies have influenced society, both in positive and negative ways. We'll read several recent news articles on these technologies and fill out a reflection sheet, then have an opportunity to decide our own opinions about these technologies as a whole.

Student Research & Reflection (15 minutes)

 **Distribute:** Digitally share the [[airesearchareas/ai-ethics/2023](#)]. This activity guide is designed to be used digitally so students can click the links and explore the online resources.

☒ **Do This:** Read over the introduction on the activity guide. Have students choose an area of specialization.

Teaching Tip

Groups: You could decide to have students work in groups for their research, with all students in the same group researching the same topic but reading different articles. This allows students to see a broader picture of the impact of their technology, especially when filling in the Research Reflection handout later.

 **Distribute:** Provide each student with one copy of the [[researchreflectionguide/ai-ethics/2023](#)] handout.

☒ **Do This:** Have students research their area, answering questions on the handout along the way. Make sure students only answer questions on the front of their reflection guide for now.

Encourage students to look over 3-4 articles from their list, making sure at least one article comes from the "Societal Impact" section and at least one article comes from the "Ethical Considerations" section. Students can also do their own research, for example: looking up the "Spotlight Person" listed in each section.

Students can skim the longer articles for key ideas, take notes, pose additional questions, and use a search engine to investigate further if time permits.

Teaching Tip

Empathize with Student Experience: Students may be personally impacted when learning about how AI can be biased. For example, students may be worried about how facial recognition bias affects them and their communities, or may have family members undergoing medical treatment and are concerned about AI in medicine. This follows advice from the A in the LEADERS framework - assume someone in the room is your datapoint.

Acknowledge that AI may impact each of us differently, and encourage students to find the proposed solutions in each of the articles, or ideate on solutions themselves. Many scholars, technologists, and policymakers are working hard to combat bias in AI.

 **Circulate:** Monitor students as they perform their research. Consider prompting students for their reactions as they read certain articles, especially on unexpected impacts of technology on society.

Regroup: Regroup the class for a class shareout

 **Prompt:** What is one thing that surprised you in your research? This could be about a positive use of this technology, or something concerning you didn't know about.

 **Discussion Goal:** This is an opportunity for students to share their thoughts and reflections from the articles, processing their thoughts and emotions. Some students may have strong reactions to the way AI is being used in society, both positively and negatively. Try to avoid open debate, but acknowledge that the pros and cons of these technologies must be weighed against the potential benefits and consequences to society. Consider reminding students that there are entire career paths dedicated towards addressing these issues in AI Ethics – policy writers, software engineers, AI ethicists, researchers, stakeholder advocates, and more.

Remarks

Any new technology can be complicated, as we've heard so far. One of the challenges is figuring out how to weigh the benefits with the harms - what potential downsides are we willing to accept in order to reap the benefits of a new technology? Or do we need to make changes so the harms are minimized? Or rethink things entirely?

This line of thinking is related to **ethics**, or guidelines on how a person or society should behave. After reading the articles, you may have an opinion on what ethical uses of this technology are. We're going to imagine what these ethical uses might be on the back of your activity guide.

☒ **Do This:** Have students flip to the back of their handout and read the introduction.

 **Display:** Display the slide with the ethical frameworks on them as well as the hypothetical scenario

Teaching Tip

Why Introduce Ethical Frameworks? This lesson introduces 5 ethical frameworks: Utilitarianism, Rights-Based Ethics, Care-Based Ethics, Equality-based, Values-Based. The goal of this lesson is *not* to become an expert on each framework, nor is it to become an expert on ethical thinking in general. Instead, the hope is students start to recognize that anytime they make a choice, that choice is influenced by covert factors on how they see and value the world **and those values have names**. For example:

- Deciding to do something because "it's what most of us want" has a name - Utilitarianism.
- Or, deciding *not* to do something because "first, do no harm" has a name - Care-Based Ethics.

By introducing students to these frameworks, we give them the language to talk about these values so they can see how they are informing the decisions they (and others) are making. And, they can also start to see other ways of thinking that might lead to different decisions that they may not have previously considered.

These are examples of **ethical frameworks**, which are different viewpoints on how to make ethical decisions. Take some consider which viewpoint resonates with your own on how to address some of the issues with the technology that you researched

☒ **Do This:** Have students fill in the back of their reflection guide

 **Circulate:** Monitor students as they complete the task, connecting their own ideas with different ethical frameworks. The frameworks are meant to help students shape their thoughts and potentially make them more coherent, and introduce them to other viewpoints on how to solve a particular problem. For example, students may default to a Utilitarian point of view where as long as the most people get the benefits everything is fine, but may shift their thinking when considering an Ethics of Care perspective and reconsider how technology impacts different groups.

Regroup: Regroup students to introduce the culminating activity of the lesson.

Teaching Tip

Final Product Options: Below are several choices for how students synthesize what they've learned in these articles and create an artifact to reflect their learning. We recommend choosing one of the options below and definitely **not** doing all of them. Or, combine or rearrange or reimagine the way you would like students to show their understanding at the end of the lesson.

Option 1: AI Code of Ethics

This section guides students towards creating a class-wide AI Code of Ethics, synthesizing what they've learned from the articles.

 **Display:** Share an example of an AI code of ethics, such as those by **Microsoft** and **Google**.

 **Prompt:** Based on what you've read and discussed, what are the most important ethical principles (rules) you believe all current and future AI must follow?

Have students discuss in their groups to brainstorm some initial ideas before regrouping to give a more specific task

☒ **Do This:** Students will write at least one principle for the class code of ethics. Ideally, their principles should be written as a sentence followed by a short paragraph that provides more context to the reader (see Google's page).

 **Circulate:** Check in with students as they complete this activity. You might ask which ethical framework most influenced their final Code of Ethics, or if they combined multiple frameworks to come up with their best practices.

Remind students that creating a code of ethics means having empathy for multiple stakeholders – the people who benefit most and benefit least from these technologies. A Code of Ethics can also include protocols for speaking to the most impacted people, or setting up ways to respond if bias occurs. The entirety of AI ethics is too large a problem to solve, but a Code of Ethics gives us some best practices.

Teaching Tip

Displaying a Class Code of Ethics: The format of your class Code of Ethics is completely up to you and your students. For example, you might ask groups to create a poster, or for each take a single slide from a shared presentation (such as the one from our [r [templateaicodeof_ethics/ai-ethics/2023](#)]), or a page from a shared document. You might even ask students to use their webcams to record short video clips that you can compile and publish as a single video. Be as creative as you wish!

Option 2: Create a Manifesto

This section guides students towards creating a personal AI Manifesto, synthesizing what they've learned from the articles.

 **Display:** Share the slide with the [Holstee Manifesto](#) on it

 **Prompt:** This is a document called the Holstee Manifesto. What do you notice about how it's structured? What is one line that stands out to you?

Give students a chance to discuss with partners before regrouping

Remarks

This is an example of a Manifesto - it's a special document created to clearly and concisely describe a set of values or beliefs. Sometimes people create manifestos as a reaction to how they see the world so it inspires change, and sometimes people create them to build community and inspire like-minded individuals. Now that we've read more about the impacts of AI on society, we're going to write our own personal manifestos about AI technologies.

 **Distribute:** Pass out a copy of the [r createamanifesto/ai-ethics/2023]. Have students read through the top section while displaying the slides on the board.

 **Display:** Display the slides describing the three reasons to create a manifesto, along with the links.

As students finish reading, open up each of the manifestos so students can see what they look like and skim some of the principles. Don't spend time deeply reading each one, but instead let students get a sense of the length of the manifesto, how it's designed visually, and the structure of each of the statements in the manifesto.

☒ **Do This:** Have students create their own personal manifesto centered on AI technology. Encourage them to be creative in how they design their manifesto.

Circulate: Check in with students as they complete this activity. Consider prompting students to tell you why they are including certain statements in their manifesto, listening for connections back to the articles from earlier in the class.

Manifestos can draw on deep emotion. Encourage students to use their passion to advocate strongly for themselves and others. They might use their manifesto as a protest sign, a call to action, a poetic description of AI failures. If a student does get particularly passionate, remind them of the work being done to combat AI bias, and encourage them to continue learning more. You may direct them to resources of different groups researching AI bias, such as the [Algorithmic Justice League](#)

(Optional) Display the Result: Consider if/how you would like to offer students the chance to share their manifestos. Some students may want to display them publicly as a call to action, while others may wish to keep theirs private and only share with you. Be sure to consider the potentially personal nature of these manifestos when deciding if/how students should share.

Option 3: Imagine Your World Art Display

This section guides students towards creating an artistic expression displaying an idealized world that uses artificial intelligence. Students also create a smaller artist-card to go along with their artistic work.

AI mistakes can be particularly harmful, and give us a pessimistic view of technology. One way to approach fear of AI is to also envision the possible benefits of responsible and fair AI systems. We want to encourage students to feel part of the solution, and to channel their creativity into imagining a better world.

 **Display:** Show the following quote from Dr. Ruha Benjamin

"Remember to imagine and craft the worlds you cannot live without, just as you dismantle the ones you cannot live within"

 **Prompt:** How do you think this quote could apply to some of the scenarios we've read about today?

Have students reflect on their own and share with a partner before returning as a full group

Remarks

This quote invites us to reimagine our circumstances and see the world as we wish it was. Based on the articles we've explored today, we're going to do exactly that - imagine a world where AI technology has been developed and implemented justly - what would that look and sound like? We're going to create a piece of artwork that imagines that world, then we'll hang it up on the walls along with a short artist card describing the world you've imagined.

 **Distribute:** Hand out the [r imagineyourworld/ai-ethics/2023] activity guide.

 **Do This:** Read the top two paragraphs and answer any questions about the directions.

Give students directions on how they should complete their artwork, for example:

- Using physical materials, such as drawing paper and paint
- Using three-dimensional materials, such as sculpting with clay or molding with cardboard
- Using a digital representation by designing using digital tools.

You may also decide to give students choice in the materials or format that they use. If teaching this in a single day, consider something on a smaller scale such as just a pencil and paper or a digital photo collage.

 **Circulate:** Monitor students as they complete their artwork. Also ensure they are completing the Artist Card to explain the artwork they are creating.

Some students may simply imagine a world where the AI problems they read about are fixed. For example, facial recognition that works on everyone's faces. Encourage students to use 'moonshot thinking' – "an approach of choosing a huge, seemingly insurmountable problem and proposing a radical solution to that problem using technology". An imagined world can be as outlandish as they want. Think big!

Display the Results: Consider how you would like to display the results in your classroom, either physically or digitally. Consider having students take a Gallery Walk to explore the artwork their peers created.

Wrap Up (5 minutes)

 **Video:** Watch the [r ethicsandaiprivacythefutureof_work/ai-ethics/2023] video as a class

 **Prompt:** How would you describe "AI ethics" to a family member or friend who didn't participate in today's lesson?

 Teaching Tip

Future Thinking: Following the lesson, students may still feel apprehensive about the impacts of AI technology on their future. They may be concerned about medical AI, college admissions, or even their own social media algorithms. If you feel comfortable, allow for students to approach you with their concerns. Remind them that

more and more experts are interested in creating responsible AI, and that students can research more advocacy groups who fight against AI injustice. You may even encourage students to write letters to local representatives about these issues.

After the Lesson

Teacher Survey

We'd love to learn more about the folks teaching these lessons and the classroom experience. Please let us know in this [**How AI Works Teacher Survey**](#).

Additional Lessons

If you'd like to teach additional lessons from the How AI Works video series, [**click here to explore additional resources**](#)

AI and Machine Learning Unit

If you'd like to dive even deeper into AI and Machine Learning, consider exploring our 5-week unit on AI and Machine Learning. [**Click here to view the unit and learn more**](#)



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Lesson 8: How AI Works In a Day

85 minutes

Overview

△ This lesson is intended for classrooms that want to show the entire **How AI Works video series** in a **single day**. It is not intended to be taught in sequence with the other lessons in this unit, which introduces each video one day at a time.

Students follow along with each video by matching vocabulary from the video, then answering a reflection question about each video. The lesson plan and slides are very sparse and open-ended to allow for improvisation and customization to fit your classroom.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- ▶ **AP** - Algorithms & Programming
- ▶ **CS** - Computing Systems
- ▶ **IC** - Impacts of Computing

Agenda

How AI Works (5 minutes)

What is Machine Learning? (10 minutes)

Training Data and Bias (10 minutes)

How Neural Networks Work (10 minutes)

How Computer Vision Works (10 minutes)

How Chatbots and Large Language Models Work (10 minutes)

Does AI Have Creativity And Imagination (10 minutes)

Ethics & AI: Equal Access and Algorithmic Bias (10 minutes)

Ethics & AI: Privacy & the Future of Work (10 minutes)

After the Lesson

Teacher Survey

Additional Lessons

AI and Machine Learning Unit

Objectives

Students will be able to:

- Explain how AI works

Preparation

- Print out the activity guide
- Project the slides on a video projector

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **How AI Works** - Slides
▼ Make a Copy
- **[Key] Vocabulary Matching** - Answer Key

For the students

- **Does AI Have Creativity and Imagination?** - Video (**Download**)
- **How AI Works: Ethics & AI: Equal Access and Algorithmic Bias** - Video
- **How AI Works: Ethics & AI: Privacy & the Future of Work** - Video
- **How AI Works: How Computer Vision Works** - Video
- **How AI Works: How Neural Networks Work** - Video
- **How AI Works: Introduction** - Video
- **How AI Works: Training Data and Bias** - Video
- **How AI Works: What is Machine Learning?** - Video
- **How Chatbots and Large Language Models Work** - Video (**Download**)

Teaching Guide

How AI Works (5 minutes)

Remarks

Today we're going to learn about AI technology! What is AI you ask? Let's find out together!

 **Video:** Show the [How AI Works: Introduction](#) video

Remarks

We're going to spend class watching this entire series! Each video will have some new vocabulary to keep track of. After each video, there will be a reflection question to answer and discuss. Let's get started!

 **Distribute:** the [Vocabulary Matching](#) notetaker to students.

What is Machine Learning? (10 minutes)

 **Video:** Watch the [How AI Works: What is Machine Learning?](#) video as a class. Students should be listening for vocabulary terms to match on their notetaker.

 **Prompt:** What's an example of AI either in your personal life or that you've seen in a movie or book?

Training Data and Bias (10 minutes)

 **Video:** Watch the [How AI Works: Training Data and Bias](#) video as a class. Students should be listening for vocabulary terms to match on their notetaker.

 **Prompt:** How does learning about human bias in machine learning change your thoughts on AI?

How Neural Networks Work (10 minutes)

 **Video:** Watch the [How AI Works: How Neural Networks Work](#) video as a class. Students should be listening for vocabulary terms to match on their notetaker.

 **Prompt:** Why is it important to have a diverse set of data when creating a recommendation system?

How Computer Vision Works (10 minutes)

 **Video:** Watch the [How AI Works: How Computer Vision Works](#) video as a class. Students should be listening for vocabulary terms to match on their notetaker.

 **Prompt:** How would you summarize how a computer "sees" something in an image?

How Chatbots and Large Language Models Work (10 minutes)

🎥 **Video:** Watch the [How Chatbots and Large Language Models Work](#) video as a class. Students should be listening for vocabulary terms to match on their notetaker.

💬 **Prompt:** Is a Large Language Model "intelligent"? Explain your answer using evidence from the video.

Does AI Have Creativity And Imagination (10 minutes)

🎥 **Video:** Watch the [Does AI Have Creativity and Imagination?](#) video as a class. Students should be listening for vocabulary terms to match on their notetaker.

💬 **Prompt:** Can AI be creative? Explain your answer using evidence from the video.

Ethics & AI: Equal Access and Algorithmic Bias (10 minutes)

🎥 **Video:** Watch the [How AI Works: Ethics & AI: Equal Access and Algorithmic Bias](#) video as a class. Students should be listening for vocabulary terms to match on their notetaker.

💬 **Prompt:** How would you describe "AI ethics" to a family member or friend?

Ethics & AI: Privacy & the Future of Work (10 minutes)

🎥 **Video:** Watch the [How AI Works: Ethics & AI: Privacy & the Future of Work](#) video as a class. Students should be listening for vocabulary terms to match on their notetaker.

💬 **Prompt:** How does AI change your thoughts on your future work and careers?

After the Lesson

Teacher Survey

We'd love to learn more about the folks teaching these lessons and the classroom experience. Please let us know in this [How AI Works Teacher Survey](#).

Additional Lessons

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